Ice Rheology Part 2: Elasticity and Viscoelasticity

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Caltech Ge 193, Spring 2026

Topics Covered

- Linear elasticity and Hooke's law for isotropic materials
- Elastic moduli and wave speeds in ice
- Maxwell viscoelasticity and the relaxation time; Kelvin–Voigt and Burgers models
- Elastic flexure of floating ice
- Examples: grounding zone tidal flexure, jökulhlaups, and supraglacial lake drainage

Primary reference: Cuffey & Paterson, *Physics of Glaciers*, 4th ed., Chapters 3 and 10; Petrenko & Whitworth, *Physics of Ice*, Chapter 4.

1 Introduction and Motivation

In the preceding lectures on ice rheology, we treated ice as a viscous fluid where stress determines the rate of deformation, and there is no memory of past states (or strain). Glen's flow law and the composite flow law of [5] describe the long-term, steady-state creep behavior of ice under sustained loading. This viscous description is appropriate for processes that unfold over timescales much longer than the material's relaxation time, a material property we define later that varies between hours and days in terrestrial glaciers.

Many important glaciological processes, however, occur on timescales short enough that ice might behave elastically. Tidal loading at grounding zones flexes ice shelves with periods of hours. Jökulhlaups drain subglacial lakes over hours to days. Supraglacial lakes load ice shelves and can drain catastrophically in minutes. On these timescales, ice can deform like an elastic solid where the Cauchy stress is proportional to strain (symmetric component of the gradient of displacement). In an elastic material, deformation is recoverable when the load is removed.

The goal of this lecture is to develop the elastic and viscoelastic constitutive relations for ice, building on the viscous framework established earlier. We begin with linear elasticity—the simplest constitutive model for a solid material—and derive Hooke's law. We then introduce the Maxwell viscoelastic model, which combines elastic and viscous behavior in a single framework and provides a natural transition between the short-timescale elastic response and the long-timescale viscous flow we have already studied. We also discuss the Kelvin–Voigt and Burgers models, which extend the Maxwell framework to capture transient (delayed elastic) creep behavior. Worked examples illustrate the application of these ideas to grounding zone flexure, jökulhlaups, and ice-shelf hydrofracture.

Two appendices derive Hooke's law and the heat equation from the first law of thermodynamics, showing how elasticity (the reversible storage of energy through molecular displacement) and heat conduction (the irreversible transport of energy through molecular vibration) arise from the same fundamental energy balance.

Plan for this lecture. We review linear elasticity and Hooke's law (§2), present the elastic properties of ice (§3), develop the elastic flexure equation for thin plates (§4), introduce Maxwell viscoelasticity (§5) along with the Kelvin–Voigt and Burgers models (§5.5), and work through three glaciological examples (§6). Appendix A derives Hooke's law from the first law of thermodynamics; Appendix B derives the standard heat equation; Appendix C derives the generalized heat equation that includes viscous dissipation and recrystallization; Appendix D derives the thin plate flexure equation from the force balance on a plate element; and Appendix E extends the plate bending theory to a Newtonian viscous fluid.

2 Linear Elasticity

2.1 Stress and Infinitesimal Strain

In the viscous framework developed in earlier lectures, the constitutive relation links deviatoric stress to the *rate* of deformation: $\dot{\varepsilon}_{ij} = f(\tau_{ij})$. In an elastic solid, the constitutive relation links Cauchy stress to the deformation itself: $\sigma_{ij} = g(\varepsilon_{ij})$.

The Cauchy stress tensor σ_{ij} and its decomposition into pressure and deviatoric parts,

$$\sigma_{ij} = -p\,\delta_{ij} + \tau_{ij}, \qquad p = -\tfrac{1}{3}\sigma_{kk}, \qquad \tau_{kk} = 0, \tag{1}$$

are the same as in the viscous theory. What changes is the kinematic quantity to which stress is related.

For small deformations, we define the **infinitesimal strain tensor**:

$$\varepsilon_{ij} = \frac{1}{2}\left(\frac{\partial s_i}{\partial x_j} + \frac{\partial s_j}{\partial x_i}\right), \tag{2}$$

where s_i is the displacement of a material point from its reference position. We reserve u_i for velocity, as in the viscous theory, so that $u_i = \dot{s}_i$ and the strain rate is $\dot{\varepsilon}_{ij} = \frac{1}{2}(\partial u_i/\partial x_j + \partial u_j/\partial x_i)$. The two frameworks are connected by a time derivative: $\dot{\varepsilon}_{ij}$ is the *instantaneous* rate of change of ε_{ij}. Strain ε_{ij} is always defined relative to the reference (zero-stress) state, whereas strain rate is the instantaneous rate of change of strain. Because strain is a state variable, it evolves over time and not necessarily at the same rate nor even monotonically, a critically important fact to remember. Much confusion can arise from forgetting that strain is relative to the reference state.

The "small deformation" assumption means $|\varepsilon_{ij}| \ll 1$, allowing us to drop higher-order terms in the full expansion and retain only the linear terms — hence, *linear* elasticity. In glaciology, elastic strains are tiny: a stress of $\tau_e = 0.1$ MPa produces an elastic strain of order $\varepsilon \sim \tau_e/\mu \sim 10^5/3.5 \times 10^9 \approx 3 \times 10^{-5}$. The small-strain approximation is therefore excellent for the elastic response of ice. (Pedantic: Linear elasticity strictly requires both small strains and a stress–strain relation that is linear and rate-independent. Ice exhibits a weak frequency dependence of its elastic moduli—a form of anelasticity—which means the moduli measured at seismic frequencies differ slightly from those at tidal frequencies. This is a departure from ideal linear elasticity, though it is a small effect at glaciological stress levels.)

2.2 Hooke's Law

For a linear elastic material, stress is a linear function of strain. The most general linear relation between two second-order tensors is

$$\boxed{\sigma_{ij} = C_{ijkl}\,\varepsilon_{kl},} \tag{3}$$

where C_{ijkl} is the **fourth-order elastic stiffness tensor**. This is Hooke's law in its most general form. The stiffness tensor has $3^4 = 81$ components, but symmetry reduces the number of independent components:

- Symmetry of the stress tensor ($\sigma_{ij} = \sigma_{ji}$) requires $C_{ijkl} = C_{jikl}$, reducing 81 to 54.
- Symmetry of the strain tensor ($\varepsilon_{kl} = \varepsilon_{lk}$) requires $C_{ijkl} = C_{ijlk}$, reducing 54 to 36.
- The existence of a strain energy function (Appendix A) requires $C_{ijkl} = C_{klij}$, reducing 36 to 21.

For a fully anisotropic crystal, 21 independent elastic constants are needed. Ice Ih has hexagonal symmetry, which reduces this to 5 independent constants. For polycrystalline ice with randomly oriented grains—the isotropic approximation—only 2 independent constants remain.

2.3 Isotropic Hooke's Law

For an **isotropic** material, the stiffness tensor must be invariant under all rotations. The most general fourth-order isotropic tensor with the required symmetries is

$$C_{ijkl} = \lambda\,\delta_{ij}\,\delta_{kl} + \mu(\delta_{ik}\,\delta_{jl} + \delta_{il}\,\delta_{jk})\,, \tag{4}$$

where λ and μ are the **Lamé parameters**. Substituting into equation (3):

$$\boxed{\sigma_{ij} = \lambda\,\varepsilon_{kk}\,\delta_{ij} + 2\mu\,\varepsilon_{ij}.} \tag{5}$$

This is Hooke's law for an isotropic solid. The parameter μ is the **shear modulus** (also denoted G), which relates shear stress to shear strain. The parameter λ is the **first Lamé parameter**, which does not have as clean a physical interpretation on its own but appears naturally in the isotropic form.

The volumetric and deviatoric parts of Hooke's law separate cleanly. Taking the trace of equation (5):

$$\sigma_{kk} = (3\lambda + 2\mu)\,\varepsilon_{kk}, \tag{6}$$

so that $p = -\frac{1}{3}(3\lambda + 2\mu)\,\varepsilon_{kk}$. The **bulk modulus** $K = \lambda + \frac{2}{3}\mu$ controls the volumetric response. The deviatoric part is

$$\tau_{ij} = 2\mu\,e_{ij}, \tag{7}$$

where $e_{ij} = \varepsilon_{ij} - \frac{1}{3}\varepsilon_{kk}\,\delta_{ij}$ is the deviatoric strain. The shear modulus μ alone controls the deviatoric (shape-changing) response.

Physical meaning of the elastic moduli. Each modulus characterizes the material's resistance to a specific type of deformation (Figure 1):

- **Shear modulus** μ (also denoted G): the ratio of shear stress to shear strain. Apply a shear stress τ_{xy} to a block of material; the block deforms into a parallelogram with shear strain $\varepsilon_{xy} = \tau_{xy}/(2\mu)$. The shear modulus controls shape change at constant volume.

- **Young's modulus** E: the ratio of axial stress to axial strain in uniaxial tension (or compression). Stretch a bar with stress σ_{xx} along x while leaving the lateral faces traction-free ($\sigma_{yy} = \sigma_{zz} = 0$); the axial strain is $\varepsilon_{xx} = \sigma_{xx}/E$. Young's modulus characterizes the stiffness felt in a simple pull test.

- **Poisson's ratio** ν: the ratio of lateral contraction to axial extension in uniaxial tension. In the same pull test, the lateral strains are $\varepsilon_{yy} = \varepsilon_{zz} = -\nu\,\varepsilon_{xx}$. A material with $\nu = 0.5$ is incompressible (no volume change); $\nu = 0$ means no lateral contraction.

- **Bulk modulus** K: the ratio of pressure to volumetric strain under uniform compression. Apply a uniform pressure p to all faces of a cube; the fractional volume change is $\varepsilon_{kk} = -p/K$. The bulk modulus controls volume change at constant shape.

Key point: two independent moduli

An isotropic elastic solid is fully characterized by two independent elastic constants. Common pairs include:

Pair	Physical interpretation
λ, μ	Lamé parameters (natural in the tensor form)
K, μ	Bulk modulus (volumetric) and shear modulus (deviatoric)
E, ν	Young's modulus (uniaxial stiffness) and Poisson's ratio (lateral contraction)

All pairs are equivalent; the choice is one of convenience. The relations are:

$$\lambda = K - \tfrac{2}{3}\mu = \frac{E\nu}{(1+\nu)(1-2\nu)}, \qquad \mu = \frac{E}{2(1+\nu)}, \qquad K = \frac{E}{3(1-2\nu)}. \tag{8}$$

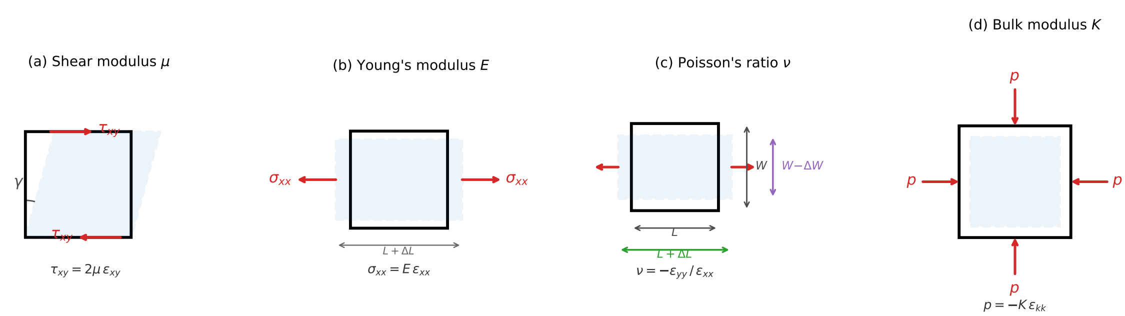


Figure 1: Schematic illustrations of the four primary elastic moduli. Solid rectangles show the undeformed reference state; dashed outlines show the deformed state. (a) Shear modulus μ: shear stress produces angular distortion at constant volume. (b) Young's modulus E: uniaxial tension produces axial extension and lateral contraction. (c) Poisson's ratio ν: the ratio of lateral contraction to axial extension in a uniaxial test. (d) Bulk modulus K: uniform pressure produces uniform volume change at constant shape.

3 Elastic Properties of Ice

3.1 Measured Elastic Moduli

The elastic moduli of polycrystalline ice Ih have been measured by a variety of techniques, including ultrasonic wave propagation, Brillouin spectroscopy, and mechanical testing. The most precise values come from wave-speed measurements, which probe the high-frequency (truly elastic) response without contamination from viscous creep. Table 1 summarizes the isotropic elastic constants for polycrystalline ice near -10°C [4].

Table 1: Isotropic elastic moduli of polycrystalline ice Ih at -10°C [4]. Values are Hill averages of the five single-crystal elastic constants, which provide the best estimate of the isotropic elastic response of a polycrystal with randomly oriented grains.

Quantity	**Symbol**	**Value**
Young's modulus	E	9.33 GPa
Shear modulus	μ	3.52 GPa
Bulk modulus	K	8.90 GPa
First Lamé parameter	λ	6.55 GPa
Poisson's ratio	ν	0.325

Several features of these values are worth noting:

1. **Ice is a stiff solid.** The shear modulus of ice (μ = 3.5 GPa) is about 25 times that of lead (0.14 GPa) and roughly 1/20 that of steel (80 GPa). The elastic response of ice is well described by the theory of linear elasticity—there is no measurable nonlinearity at glaciological stress levels.

2. **Poisson's ratio is close to 1/3.** A Poisson's ratio $\nu = 0.325$ means that ice contracts laterally by about one-third of the amount it extends axially under uniaxial tension. For comparison, $\nu = 0.5$ corresponds to an incompressible material (no volume change under any deformation), and $\nu = 0$ corresponds to a material that does not contract laterally at all (like cork).

3. **Temperature dependence is weak.** The elastic moduli increase by roughly 5% between 0°C and -40°C [4]. This is a minor effect compared to the strong temperature dependence of viscous creep (activation energies of 49–180 $\mathrm{kJ\,mol^{-1}}$). For most glaciological calculations, the elastic moduli can be treated as constants.

4. **The shear modulus has a weak frequency dependence.** The values in Table 1 are measured at ultrasonic frequencies (∼MHz) and represent the unrelaxed (instantaneous) elastic response. At lower frequencies—tidal ($\sim 10^{-5}$ Hz) or seismic (∼1–100 Hz)—the effective shear modulus is slightly lower because thermally activated processes at grain boundaries and lattice defects allow a small amount of anelastic relaxation on the timescale of each loading cycle. Physically, this arises because grain boundaries in polycrystalline ice can slide viscously over short distances (grain boundary sliding) and dislocations can bow between pinning points, both of which accommodate a small additional strain that is recoverable but delayed. At high frequencies, these relaxation mechanisms cannot keep pace with the loading and the material responds with the full lattice stiffness; at lower frequencies, the mechanisms contribute additional compliance. The effect is small—the shear modulus decreases by roughly 5–10% from MHz to tidal frequencies—but it means that the "elastic modulus" inferred from tidal flexure observations is systematically lower than the laboratory ultrasonic value, even

before accounting for viscous creep (which further reduces the effective modulus through the viscoelastic mechanism described in §5).

3.2 Elastic Wave Speeds

The elastic moduli determine the speeds of seismic waves in ice. Substituting Hooke's law (5) into Cauchy's equation of motion ($\sigma_{ij,j} + f_i = \rho\,\ddot{s}_i$, in the absence of body forces) yields the Navier–Cauchy equation:

$$(\lambda + \mu)\,\frac{\partial^2 s_j}{\partial x_i \partial x_j} + \mu\,\frac{\partial^2 s_i}{\partial x_j \partial x_j} = \rho\,\frac{\partial^2 s_i}{\partial t^2}. \tag{9}$$

This wave equation admits two types of plane-wave solutions:

- **P-waves** (compressional): $v_p = \sqrt{(\lambda + 2\mu)/\rho} = \sqrt{(K + \frac{4}{3}\mu)/\rho}$,
- **S-waves** (shear): $v_s = \sqrt{\mu/\rho}$.

For ice ($\rho_i = 917$ kg m^{-3}):

$$v_p = \sqrt{\frac{6550 + 2 \times 3520}{917} \times 10^6} \approx 3850\ \mathrm{m\,s^{-1}}, \qquad v_s = \sqrt{\frac{3520 \times 10^6}{917}} \approx 1960\ \mathrm{m\,s^{-1}}. \tag{10}$$

These values agree well with seismic observations in glaciers and ice sheets [7]. The P-wave speed in ice is notably higher than in water (1480 m s^{-1}) and lower than in most rocks (4000–7000 m s^{-1}).

4 Elastic Flexure of Thin Plates

Many glaciological applications of elasticity involve the bending of ice shelves and glacier tongues—structures that are thin compared to their horizontal extent. The theory of thin elastic plates provides an efficient framework for these problems.

4.1 The Flexure Equation

Consider a thin plate of thickness h and flexural rigidity

$$D = \frac{E\,h^3}{12(1 - \nu^2)}, \tag{11}$$

resting on or floating in a fluid of density ρ_w. Let $w(x, y)$ denote the vertical deflection of the plate from its equilibrium position. For small deflections ($|w| \ll h$), the governing equation is

$$\boxed{D\,\nabla^4 w + \rho_w\,g\,w = q(x, y),} \tag{12}$$

where $\nabla^4 = \partial^4/\partial x^4 + 2\,\partial^4/(\partial x^2\,\partial y^2) + \partial^4/\partial y^4$ is the biharmonic operator, $\rho_w\,g\,w$ is the buoyancy restoring force per unit area (for a floating plate), and $q(x, y)$ is any applied load per unit area. Appendix D derives this equation from the force and moment balance on a plate element.

In one horizontal dimension (e.g., a beam or a plate that varies only in x):

$$D\,\frac{d^4 w}{dx^4} + \rho_w\,g\,w = q(x). \tag{13}$$

The homogeneous solution (with $q = 0$) involves a characteristic length scale, the **flexural wavelength**:

$$\boxed{\ell_f = \left(\frac{4D}{\rho_w\, g}\right)^{1/4} = \left(\frac{E\, h^3}{3(1-\nu^2)\,\rho_w\, g}\right)^{1/4}.} \tag{14}$$

The flexural wavelength sets the horizontal distance over which the plate adjusts from a loaded to an unloaded state. Features much smaller than ℓ_f are supported by the plate's rigidity; features much larger than ℓ_f are in isostatic equilibrium with the underlying fluid.

4.2 Flexural Wavelength of Ice Shelves

For an ice shelf of thickness h, using $E = 9.33$ GPa, $\nu = 0.325$, and $\rho_w = 1025\ \mathrm{kg\,m^{-3}}$:

$$\ell_f = \left(\frac{9.33 \times 10^9 \times h^3}{3 \times 0.894 \times 1025 \times 9.8}\right)^{1/4} = \left(\frac{9.33 \times 10^9}{2.70 \times 10^4}\right)^{1/4} h^{3/4} \approx 24.2\, h^{3/4}, \tag{15}$$

where h is in meters and ℓ_f is in meters. Representative values:

Ice thickness h (m)	Flexural rigidity D (N m)	Flexural wavelength ℓ_f (km)
200	6.9×10^{15}	1.3
500	1.1×10^{17}	2.6
1000	8.7×10^{17}	4.3

The flexural wavelength is typically 1–5 km for Antarctic ice shelves. This length scale appears repeatedly in our worked examples.

4.3 Bending Stresses

The bending of a thin plate produces normal stresses that vary linearly through the thickness:

$$\sigma_{xx}(z) = -\frac{E\, z}{1-\nu^2}\left(\frac{\partial^2 w}{\partial x^2} + \nu\,\frac{\partial^2 w}{\partial y^2}\right), \tag{16}$$

where z is measured from the plate's neutral surface (midplane), with $z > 0$ upward. The maximum tensile stress occurs at the surface farthest from the center of curvature. For a plate deflecting downward ($w < 0$, concave upward), the maximum tensile stress is at the bottom surface ($z = -h/2$). For a plate deflecting upward, the maximum tensile stress is at the top surface.

In one horizontal dimension:

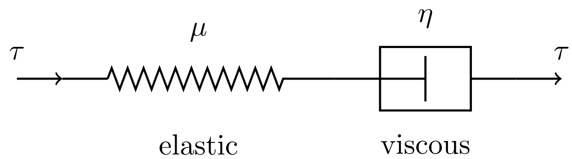
$$\sigma_{xx}^{\max} = \frac{E\, h}{2(1-\nu^2)}\left|\frac{d^2 w}{dx^2}\right| = \frac{6\,|M|}{h^2}, \tag{17}$$

where $M = -D\, d^2w/dx^2$ is the bending moment per unit width.

5 Maxwell Viscoelasticity

5.1 The Maxwell Model

The creep flow laws developed in the preceding lecture describe the long-timescale behavior of ice under sustained loading. The elastic theory of §2 describes the instantaneous response to a change in load. To bridge these regimes, we need a constitutive model that includes both elastic and viscous behavior. The simplest such model is the **Maxwell model**, which places an elastic element (spring) and a viscous element (dashpot) in series.



In a series arrangement, both elements carry the same stress, and the total strain rate is the sum of the individual strain rates:

$$\dot{\varepsilon}_{ij} = \dot{\varepsilon}_{ij}^{(\text{elastic})} + \dot{\varepsilon}_{ij}^{(\text{viscous})}. \tag{18}$$

The elastic strain rate is the time derivative of the elastic constitutive relation (7):

$$\dot{\varepsilon}_{ij}^{(\text{elastic})} = \frac{1}{2\mu}\,\dot{\tau}_{ij}. \tag{19}$$

assuming incompressibility. The viscous strain rate is given by the creep flow law. For Glen's law:

$$\dot{\varepsilon}_{ij}^{(\text{viscous})} = A\,\tau_e^{n-1}\,\tau_{ij}. \tag{20}$$

Combining:

$$\boxed{\dot{\varepsilon}_{ij} = \frac{1}{2\mu}\,\dot{\tau}_{ij} + A\,\tau_e^{n-1}\,\tau_{ij}.} \tag{21}$$

This is the Maxwell viscoelastic constitutive equation for ice. The first term describes the instantaneous elastic response to changes in stress; the second describes the ongoing viscous creep under sustained stress.

Equation (21) is written for the deviatoric part of the stress–strain relation. For the volumetric part, we assume a purely elastic response:

$$\sigma_{kk} = 3K\,\varepsilon_{kk}, \tag{22}$$

because ice does not undergo viscous volume change (the creep mechanisms are isochoric).

5.2 The Maxwell Relaxation Time

The Maxwell model has a characteristic timescale, the **relaxation time**, that separates elastic-dominated from viscous-dominated behavior. To find it, consider the 1D version of equation (21) and suppose the total strain is held fixed ($\dot{\varepsilon} = 0$) after an instantaneous loading. Then

$$\frac{1}{2\mu}\,\dot{\tau} + \frac{\tau}{2\eta} = 0, \tag{23}$$

where η is the effective viscosity, related to Glen's law by $\eta = \tau_e/(2\dot{\varepsilon}_e) = 1/(2A\,\tau_e^{n-1})$. For constant η (i.e., Newtonian viscosity or linearization about a given stress), the solution is exponential decay:

$$\tau(t) = \tau_0\,\exp(-t/t_M)\,, \qquad t_M = \frac{\eta}{\mu}. \tag{24}$$

The **Maxwell time** $t_M = \eta/\mu$ is the e-folding time for stress relaxation at constant strain. It has a direct physical meaning: elastic stresses decay on a timescale t_M as the viscous element takes up the deformation.

For ice with Glen's law ($n = 3$), the effective viscosity $\eta = 1/(2A\,\tau_e^2)$ depends on the stress level, so the Maxwell time is also stress-dependent:

$$\boxed{t_M = \frac{1}{2\mu\, A\, \tau_e^{n-1}}.} \tag{25}$$

The Maxwell time is short where ice is warm and highly stressed, and long where ice is cold and under low stress. Table 2 gives representative values.

Table 2: Maxwell relaxation time t_M for polycrystalline ice under different conditions, computed from equation (25) using $\mu = 3.5$ GPa, $n = 3$, and values of A from the Arrhenius relation with $Q = 90$ kJ mol^{-1} (below 263 K) and $Q = 139$ kJ mol^{-1} (above 263 K), fitted to the tabulated values of [2].

Setting	T (K)	τ_e (MPa)	A (Pa^{-3} s^{-1})	t_M
Warm shear margin	270	0.2	2.0×10^{-24}	30 min
Warm grounding zone	270	0.1	2.0×10^{-24}	2.0 hours
Temperate glacier	263	0.1	3.5×10^{-25}	11 hours
Cold ice-sheet interior	243	0.05	1.2×10^{-26}	55 days
Cold, low-stress interior	233	0.01	1.6×10^{-27}	$\sim$28 years

The Maxwell time ranges from about half an hour in warm, highly stressed ice (shear margins, grounding zones) to decades in the cold, low-stress interiors of ice sheets. This enormous range—more than five orders of magnitude—means that the same material can behave as an elastic solid or a viscous fluid depending on the local conditions and the timescale of the forcing.

5.3 The Deborah Number

A useful dimensionless number for deciding whether a process is elastic, viscous, or viscoelastic is the **Deborah number**:

$$\mathrm{De} = \frac{t_M}{t_p}, \tag{26}$$

where t_p is the characteristic timescale of the process or forcing.

- $\mathrm{De} \gg 1$: the relaxation time is much longer than the process timescale. The viscous element has no time to respond, and the material behaves elastically.
- $\mathrm{De} \ll 1$: the relaxation time is much shorter than the process timescale. Elastic stresses relax away quickly, and the material behaves as a viscous fluid.
- $\mathrm{De} \sim 1$: both elastic and viscous effects are important, and the full Maxwell constitutive equation must be used.

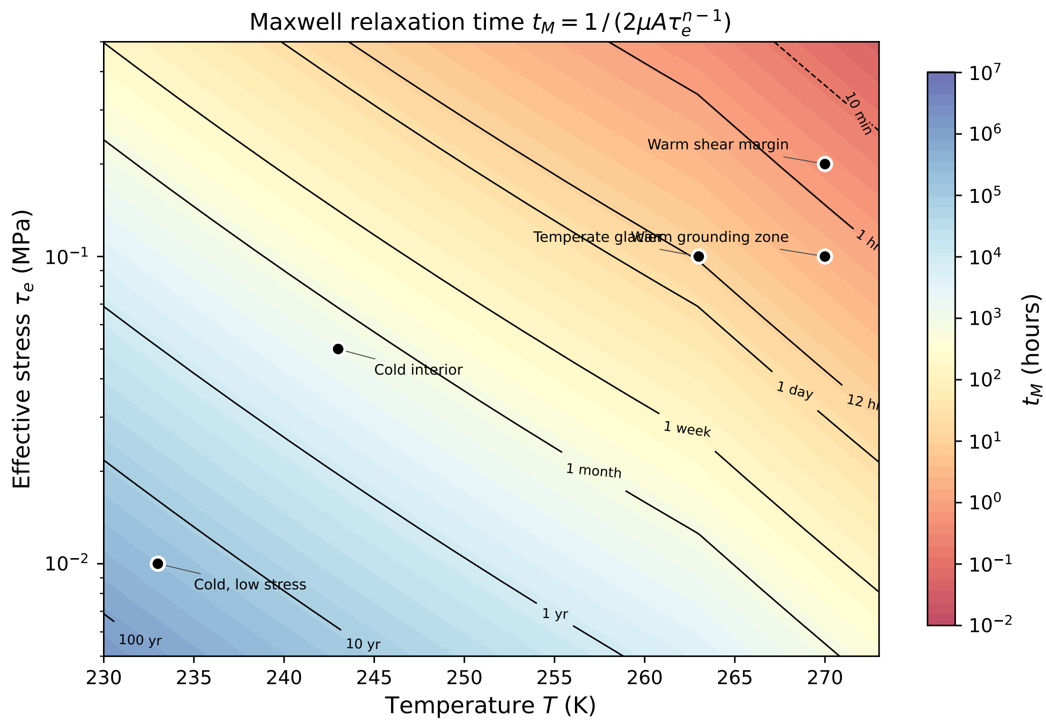


Figure 2: Maxwell relaxation time $t_M = 1/(2\mu A\,\tau_e^{n-1})$ as a function of temperature and effective stress, computed using the Arrhenius relation for $A(T)$. Contour labels indicate the relaxation time. Black dots mark the five settings from Table 2.

Key point: when does elasticity matter?

Elasticity matters when the Deborah number is of order unity or larger. For glacier ice, this means:

- **Tidal forcing** ($t_p \sim 12$ hours): De $\sim$ 0.04–0.9 in warm ice near grounding zones. Viscoelastic effects are important.
- **Jökulhlaup drainage** ($t_p \sim$ hours to days): De $\sim$ 0.05–5 depending on conditions. Both elastic and viscous responses are relevant.
- **Lake drainage / hydrofracture** ($t_p \sim$ minutes to hours): De $\gg 1$ everywhere. The response is essentially elastic.
- **Seismic waves** ($t_p \sim 0.01$–1 s): De $\gg 1$ everywhere. Purely elastic.
- **Seasonal flow variations** ($t_p \sim$ months): De $\ll 1$ in warm ice. Viscous.
- **Ice-sheet evolution** ($t_p \sim 10^3$–10^5 years): De $\ll 1$. Purely viscous.

The line between "elastic" and "viscous" behavior is not a property of the material alone—it depends on both the material state (which controls t_M) and the timescale of the process (which sets t_p).

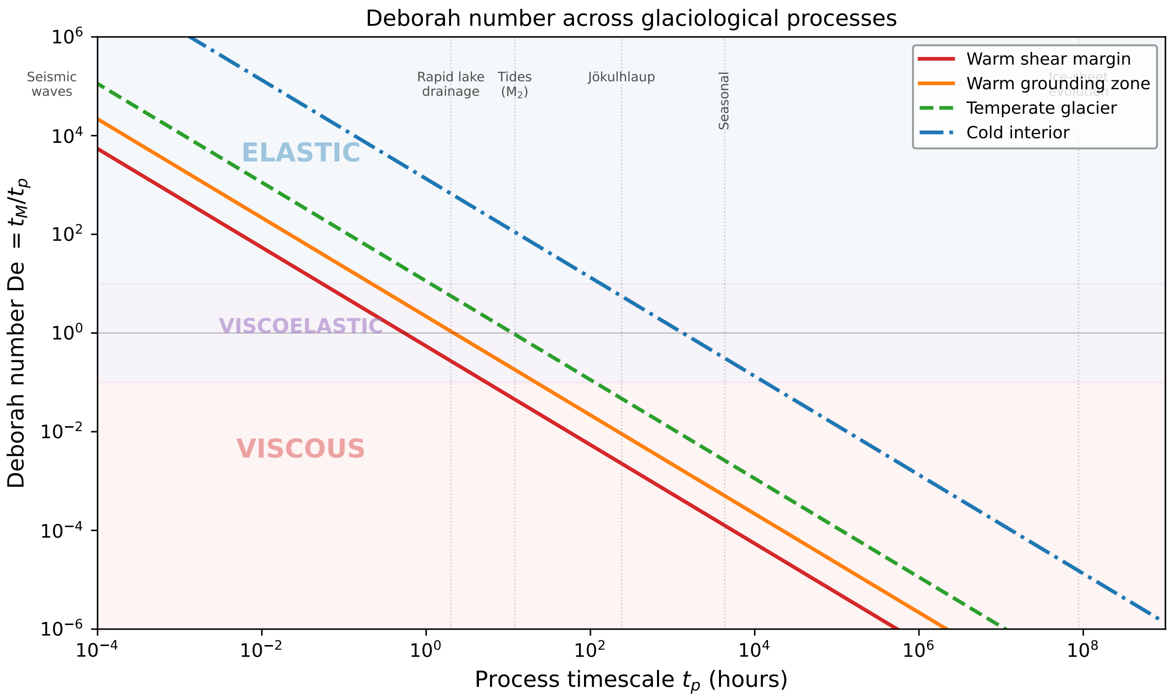


Figure 3: Deborah number De $= t_M/t_p$ as a function of process timescale for four glaciological settings (Table 2). Vertical dotted lines mark characteristic timescales of glaciological processes. The shaded bands indicate the elastic (De $\gg 1$), viscoelastic (De ~ 1), and viscous (De $\ll 1$) regimes.

5.4 Response to Step Loading

The response of a Maxwell material to an instantaneous step load reveals the interplay between elastic and viscous deformation.

Step stress (creep test). Apply a constant stress τ_0 at $t = 0$. With $\dot{\tau} = 0$ for $t > 0$, equation (21) gives:

$$\varepsilon(t) = \frac{\tau_0}{2\mu} + \frac{\tau_0}{2\eta}\, t = \frac{\tau_0}{2\mu}\left(1 + \frac{t}{t_M}\right). \tag{27}$$

There is an instantaneous elastic strain $\tau_0/(2\mu)$ at $t = 0$, followed by a steady viscous creep at rate $\tau_0/(2\eta)$. At $t = t_M$, the viscous strain equals the elastic strain. For $t \gg t_M$, the elastic contribution is negligible and the response is purely viscous.

Step strain (relaxation test). Apply an instantaneous strain ε_0 at $t = 0$ and hold it constant ($\dot{\varepsilon} = 0$). From equation (24):

$$\tau(t) = 2\mu\, \varepsilon_0 \exp(-t/t_M)\,. \tag{28}$$

The initial stress is $2\mu\,\varepsilon_0$ (fully elastic), and it relaxes exponentially as the viscous element accommodates the imposed strain. After a time t_M, the stress has decayed to $1/e \approx 37\%$ of its initial value. After $3\,t_M$, it has decayed to $\sim$5%.

Oscillatory loading (harmonic forcing). For a sinusoidal stress $\tau(t) = \tau_0 \sin(\omega t)$, the strain response of a linear Maxwell material is obtained by direct integration of the constitutive equation (21) (with $\dot{\tau} = \tau_0\,\omega\cos(\omega t)$ and $\tau/(2\eta) = \tau_0\sin(\omega t)/(2\eta)$):

$$\varepsilon(t) = \frac{\tau_0}{2\mu}\,\sin(\omega t) + \frac{\tau_0}{2\eta\,\omega}\left[1 - \cos(\omega t)\right]. \tag{29}$$

The first term is the elastic strain (in phase with the stress); the second is the viscous contribution, which oscillates about a mean offset of $\tau_0/(2\eta\omega)$. At high frequency ($\omega\,t_M \gg 1$), the elastic term dominates and the strain amplitude approaches $\tau_0/(2\mu)$ (elastic limit). At low frequency ($\omega\,t_M \ll 1$), the viscous term dominates and the strain amplitude scales as $\tau_0/(2\eta\omega) = \tau_0/(2\mu\,\omega\,t_M)$.

The tidal flexure of ice shelves near grounding zones is a natural example of oscillatory loading. The tidal period ($\sim$12 hours for the dominant M_2 constituent) is comparable to the Maxwell time of warm ice near the grounding line, placing this problem squarely in the viscoelastic regime.

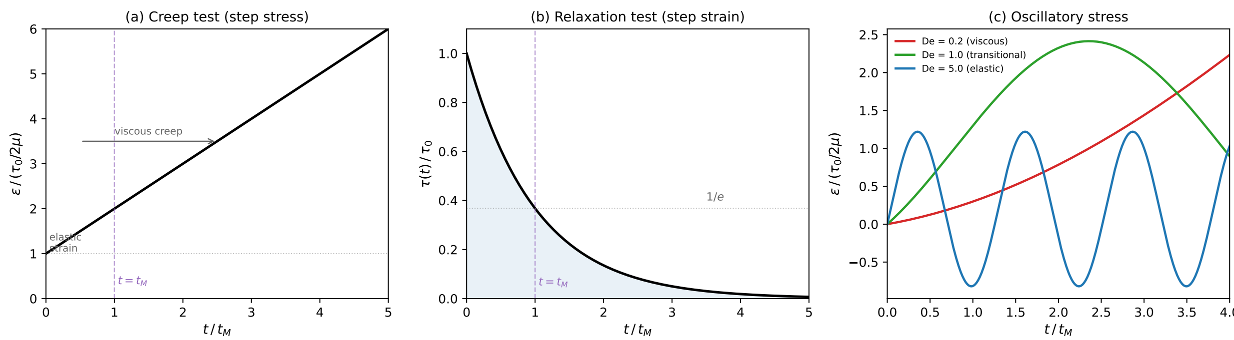


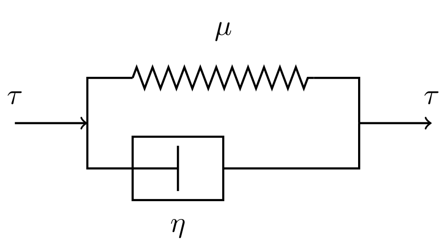
Figure 4: Response of a linear Maxwell material to three types of loading. (a) Creep test: a step stress produces an instantaneous elastic strain followed by steady viscous creep; at $t = t_M$ the viscous strain equals the elastic strain. (b) Relaxation test: a step strain produces an initial elastic stress that decays exponentially with e-folding time t_M. (c) Oscillatory stress at three Deborah numbers: for De $= 0.2$ (viscous regime) the strain drifts; for De $= 5$ (elastic regime) the strain oscillates in phase with the stress; for De $= 1$ the response is transitional.

5.5 Other Viscoelastic Models: Kelvin–Voigt and Burgers

The Maxwell model is not the only way to combine elastic and viscous elements. Two other classical models—Kelvin–Voigt and Burgers—arise from different arrangements and provide qualitatively different mechanical behavior.

5.5.1 The Kelvin–Voigt Model

The **Kelvin–Voigt model** places a spring and dashpot in *parallel*, so that both elements share the same strain but carry different fractions of the stress:



The constitutive equation is:

$$\tau_{ij} = 2\mu\,\varepsilon_{ij} + 2\eta\,\dot{\varepsilon}_{ij}. \tag{30}$$

In a creep test (constant applied stress τ_0), the 1D response is:

$$\varepsilon(t) = \frac{\tau_0}{2\mu}\left(1 - e^{-t/t_R}\right), \qquad t_R = \frac{\eta}{\mu}, \tag{31}$$

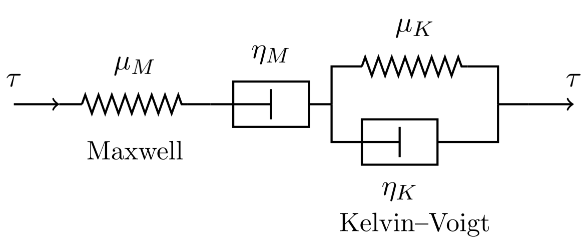
where t_R is the **retardation time**. The strain asymptotes exponentially to the elastic equilibrium value $\tau_0/(2\mu)$; on removal of the load, the strain recovers completely on the same timescale. This is the key distinction from the Maxwell model: the Kelvin–Voigt material exhibits *delayed elasticity*—all deformation is eventually recoverable—whereas the Maxwell material exhibits permanent viscous flow.

In a relaxation test (constant applied strain), the Kelvin–Voigt model predicts an instantaneously infinite stress (the dashpot resists any instantaneous deformation), which then immediately drops to the finite elastic value $2\mu\,\varepsilon_0$. The model therefore cannot accommodate instantaneous elastic strain—a significant limitation.

Glaciological relevance. The Kelvin–Voigt model, used alone, does not capture the two behaviors most important for glacier dynamics: instantaneous elastic response and long-term irreversible creep. For this reason, the standalone Kelvin–Voigt model is not used in glacier dynamics to the author's knowledge. Its value lies in its role as a building block for more complete models, particularly the Burgers model described next.

5.5.2 The Burgers Model

The **Burgers model** combines a Maxwell element in series with a Kelvin–Voigt element, giving four parameters: two elastic moduli (μ_M, μ_K) and two viscosities (η_M, η_K).



Because the elements are in series, the total strain rate is the sum of the Maxwell and Kelvin–Voigt strain rates. For a creep test at constant stress τ_0, the 1D strain response is:

$$\boxed{\varepsilon(t) = \underbrace{\frac{\tau_0}{2\mu_M}}_{\text{instantaneous elastic}} + \underbrace{\frac{\tau_0}{2\eta_M}\,t}_{\text{steady-state creep}} + \underbrace{\frac{\tau_0}{2\mu_K}\left(1 - e^{-t/t_K}\right)}_{\text{delayed elastic (transient)}}}\,, \tag{32}$$

where $t_K = \eta_K/\mu_K$ is the retardation time of the Kelvin–Voigt element. The three terms correspond to three distinct physical regimes:

1. An **instantaneous elastic** strain at $t = 0$ (from the Maxwell spring μ_M).
2. A **transient creep** phase that saturates on timescale t_K (from the Kelvin–Voigt unit), representing delayed elastic recovery associated with reversible processes such as grain-boundary sliding accommodated by elastic back-stresses.

3. A **steady-state viscous creep** at long times (from the Maxwell dashpot η_M), representing permanent dislocation creep.

These three regimes map directly onto the primary, secondary, and tertiary stages of a laboratory creep test, making the Burgers model an attractive phenomenological description of the full transient response of polycrystalline ice.

Characteristic timescales. The Burgers model has two internal timescales: the Maxwell relaxation time $t_M = \eta_M/\mu_M$, which controls stress relaxation at constant strain, and the Kelvin–Voigt retardation time $t_K = \eta_K/\mu_K$, which controls the duration of transient creep. For ice, laboratory experiments and field observations suggest $t_K \sim$ hours to days—comparable to the Maxwell time in many glaciological settings—while the steady-state viscosity η_M corresponds to the effective viscosity of Glen's law [2].

Glaciological applications. The Burgers model has been used to interpret:

- Tidal modulation of ice-stream velocity, where the transient (delayed elastic) response produces a velocity signal at the tidal frequency that decays with distance from the grounding line [6, 10].
- Short-term transient creep of ice observed in laboratory experiments and borehole deformation measurements, where the delayed elastic recovery upon unloading is well described by the Kelvin–Voigt element [2].
- Laboratory creep experiments at stresses below the Glen's-law regime, where the transient phase is pronounced and the Burgers model provides a better fit than the Maxwell model alone [2].

5.5.3 Comparison of the Three Models

Table 3 summarizes the key features of the three viscoelastic models.

Table 3: Comparison of linear viscoelastic models. "Instant elastic" indicates whether the model permits an instantaneous elastic strain; "permanent flow" indicates whether strain accumulates indefinitely under constant stress; "recoverable" indicates whether strain is fully recovered on unloading.

Model	**Instant elastic**	**Permanent flow**	**Recoverable**	**Use in glaciology**
Maxwell	Yes	Yes	No	Most common
Kelvin–Voigt	No	No	Yes	Not used alone
Burgers	Yes	Yes	Partially	Occasional

The Maxwell model is the standard choice for most problems in glacier dynamics because it captures the two most important behaviors: instantaneous elastic response to rapid loading and permanent viscous flow under sustained stress. The Burgers model is preferred when the transient creep phase is important—particularly for tidal problems where the forcing period is comparable to the retardation time t_K, so that the delayed elastic response contributes significantly to the observed deformation.

6 Worked Examples

6.1 Example 1: Grounding Zone Tidal Flexure

The grounding zone is the transition between grounded ice (resting on bedrock) and floating ice (an ice shelf or glacier tongue). Ocean tides cause the floating ice to rise and fall, while the grounded ice remains fixed. The transition between these two states involves flexure of the ice shelf over a horizontal distance comparable to the flexural wavelength.

Setup. Consider a one-dimensional cross section perpendicular to the grounding line. Let x denote distance from the grounding line, with $x > 0$ in the floating ice shelf and $x < 0$ in the grounded ice. The ice shelf has uniform thickness h and floats on the ocean with a tidally varying water level. The tidal displacement at $x \to \infty$ is $w_0(t) = w_0 \sin(\omega t)$, where w_0 is the tidal amplitude. At the grounding line ($x = 0$), the ice is clamped: $w(0, t) = 0$ and $dw/dx|_{x=0} = 0$ (no deflection, no slope).

Elastic solution. The flexure equation (13) with $q = 0$ has the general solution

$$w(x) = e^{-x/\ell_f}\left[C_1 \cos\left(\frac{x}{\ell_f}\right) + C_2 \sin\left(\frac{x}{\ell_f}\right)\right] + e^{x/\ell_f}\left[C_3 \cos\left(\frac{x}{\ell_f}\right) + C_4 \sin\left(\frac{x}{\ell_f}\right)\right], \tag{33}$$

where $\ell_f = (4D/\rho_w g)^{1/4}$. Requiring boundedness as $x \to \infty$ sets $C_3 = C_4 = 0$. Far from the grounding line, the ice shelf is in hydrostatic equilibrium and moves with the tide: $w(x \to \infty) = w_0$. The solution satisfying the boundary conditions $w(0) = 0$ and $w'(0) = 0$ is

$$\boxed{w(x, t) = w_0 \sin(\omega t)\left[1 - e^{-x/\ell_f}\left(\cos\frac{x}{\ell_f} + \sin\frac{x}{\ell_f}\right)\right].} \tag{34}$$

Derivation: tidal flexure profile

The general bounded solution for $x > 0$ is

$$w(x) = w_0 + e^{-x/\ell_f}\left[C_1 \cos\frac{x}{\ell_f} + C_2 \sin\frac{x}{\ell_f}\right],$$

where we have added the far-field displacement w_0 as a particular solution (the weight of the tide is supported by buoyancy, not by the plate). Applying $w(0) = 0$: $C_1 = -w_0$. For the slope condition, write $\xi = x/\ell_f$:

$$w(\xi) = w_0 + e^{-\xi}[C_1 \cos\xi + C_2 \sin\xi].$$

Differentiate:

$$w'(\xi) = e^{-\xi}[(-C_1 + C_2)\cos\xi + (-C_2 - C_1)\sin\xi].$$

At $\xi = 0$: $w'(0) = -C_1 + C_2 = 0$, so $C_2 = C_1 = -w_0$. Substituting:

$$w(x) = w_0\left[1 - e^{-x/\ell_f}\left(\cos\frac{x}{\ell_f} + \sin\frac{x}{\ell_f}\right)\right]. \qquad \square$$

Bending stresses. The maximum bending stress occurs at the grounding line ($x = 0$), where the curvature is largest. From equation (34):

$$\left.\frac{d^2 w}{dx^2}\right|_{x=0} = \frac{2\,w_0}{\ell_f^2}. \tag{35}$$

The maximum tensile stress at the ice surface is (from equation (17)):

$$\sigma_{\max} = \frac{E\,h}{2(1-\nu^2)}\,\frac{2\,w_0}{\ell_f^2} = \frac{E\,h\,w_0}{(1-\nu^2)\,\ell_f^2}. \tag{36}$$

Numerical evaluation. For an ice shelf with $h = 500$ m and a tidal amplitude $w_0 = 1$ m:

- Flexural rigidity: $D = 9.33 \times 10^9 \times 500^3/[12 \times 0.894] = 1.09 \times 10^{17}$ N m.
- Flexural wavelength: $\ell_f = (4 \times 1.09 \times 10^{17}/(1025 \times 9.8))^{1/4} = 2.6$ km.
- Maximum bending stress: $\sigma_{\max} = 9.33 \times 10^9 \times 500 \times 1/(0.894 \times (2600)^2) = 770$ kPa.

This stress is below but approaching the tensile strength of ice (∼1 MPa). Tidal flexure alone does not typically fracture intact ice, but the stresses are large enough to promote fatigue crack growth in pre-existing crevasses at grounding zones, a process that has been linked to ice-shelf rifting.

Viscoelastic correction. The purely elastic solution (34) assumes that the ice responds instantaneously and completely to each tidal cycle. At the grounding zone of a warm ice shelf, the Maxwell time ($t_M \sim 2$ hours; Table 2) is comparable to the tidal period (∼12 hours for M_2), so the Deborah number De ≈ 0.17. The viscous component therefore accommodates a significant fraction of the tidal flexure. This has two consequences:

1. The effective flexural rigidity is reduced, because part of the deformation is viscous and unrecoverable. The flexural wavelength is shorter than the elastic prediction.
2. There is a phase lag between the tidal forcing and the ice-shelf response, because the viscous element introduces a delay (Appendix E, equation (108)).

Both effects are observed in GPS and tiltmeter measurements at Antarctic grounding zones [14, 9]. Elgart et al. [3] used ICESat-2 laser altimetry to estimate the effective elastic modulus in grounding zones of the Ross Ice Shelf from tidal flexure profiles, finding that quantitative agreement with observed flexure requires accounting for viscoelastic effects through a reduced effective modulus.

6.2 Example 2: Jökulhlaups and Elastic Rebound

A jökulhlaup is a glacial outburst flood caused by the sudden drainage of a subglacial or ice-marginal lake. The drainage removes a distributed load from the glacier surface or bed, and the ice responds to this unloading. Whether the response is elastic, viscous, or viscoelastic depends on the drainage timescale relative to the Maxwell time.

Setup. Consider a subglacial lake of area $\mathcal{A}$ and depth d_ℓ beneath an ice sheet of thickness h. The lake supports a fraction of the overburden pressure: the water pressure at the ice–lake interface is $p_w = \rho_w\,g\,d_\ell$. When the lake drains on a timescale t_d, this hydraulic support is removed and the ice above the former lake must be supported entirely by the surrounding ice and bed.

Elastic limit: fast drainage. If $t_d \ll t_M$, the ice responds elastically. The removal of the water load $q = \rho_w\,g\,d_\ell$ over the lake area produces an elastic subsidence of the ice surface. For a circular lake of radius R:

- If $R \ll \ell_f$: the plate is effectively rigid over the lake area, and the deflection is controlled by the plate's flexural resistance. The maximum subsidence scales as $w \sim q\,R^4/D$.

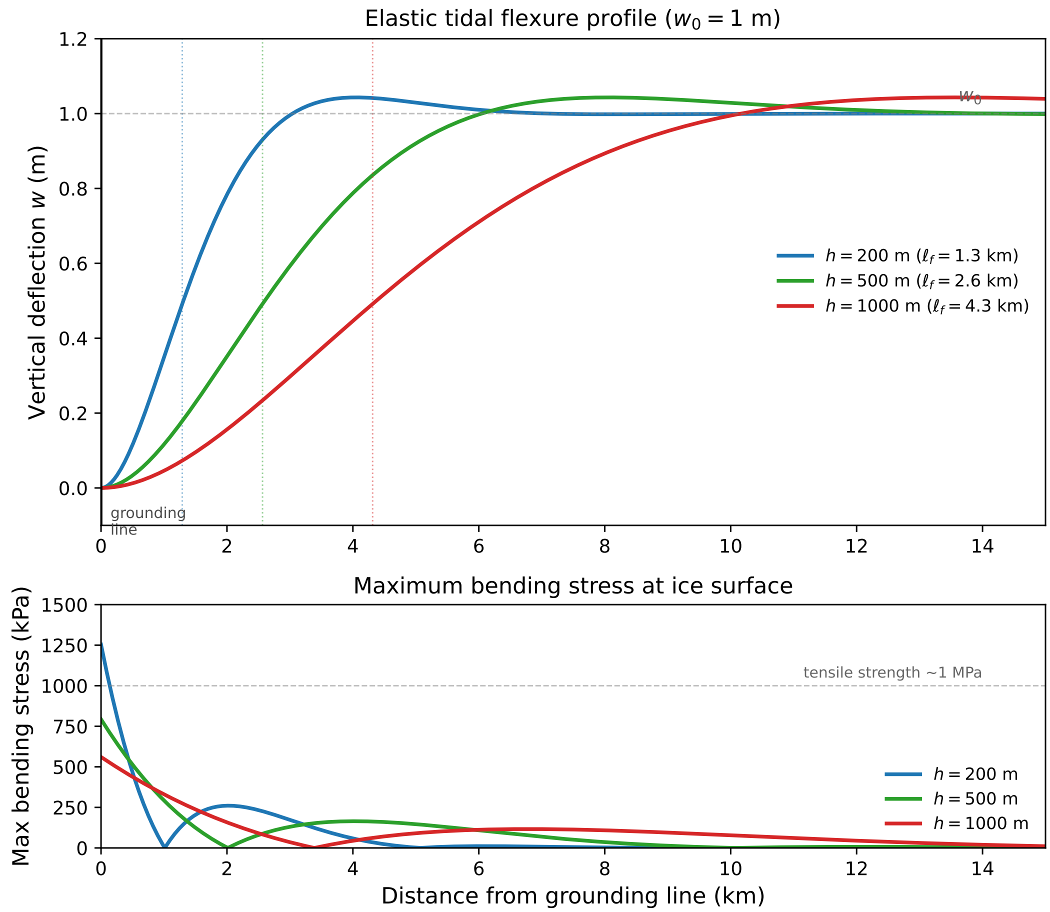


Figure 5: Elastic tidal flexure at a grounding zone for a tidal amplitude $w_0 = 1$ m and three ice thicknesses. Top: vertical deflection profile $w(x)$ from equation (34), with the flexural wavelength ℓ_f marked for each thickness. Bottom: maximum bending stress at the ice surface, showing that stresses approach but remain below the tensile strength of intact ice ($\sim$1 MPa) for thick ice shelves.

- If $R \gg \ell_f$: the center of the former lake reaches isostatic equilibrium while the edges are supported by flexure.

For a more instructive estimate, consider the Deborah number. Suppose the ice above the lake is at $T = 270$ K with $\tau_e \approx 0.1$ MPa. From Table 2, $t_M \approx 2$ hours.

Case 1: Grímsvötn-type jökulhlaup ($t_d \sim$ 1–2 weeks). The Deborah number is De $= t_M/t_d \approx 2/(24 \times 10) \approx 0.008$. The drainage is slow compared to the Maxwell time, so the ice responds viscously. Elastic stresses relax as the lake drains, and the surface subsidence is controlled by viscous flow. GPS observations of surface displacement show smooth, gradual subsidence consistent with a viscous response.

Case 2: Rapid subglacial lake drainage ($t_d \sim 2$ hours). De $= t_M/t_d \approx 1.0$. The drainage timescale is comparable to the Maxwell time. The initial surface response is elastic—a rapid subsidence of order centimeters to decimeters—followed by viscous relaxation over subsequent hours to days. Both elastic and viscous effects are important, and the full Maxwell model is needed to interpret surface displacement observations.

Elastic subsidence estimate. For an instantaneous drainage of a subglacial lake with $d_\ell = 10$ m and $R = 5$ km beneath $h = 1000$ m of ice:

$$q = \rho_w \, g \, d_\ell = 1000 \times 9.8 \times 10 = 98 \text{ kPa}, \tag{37}$$

and the flexural wavelength is $\ell_f \approx 4.3$ km (for $h = 1000$ m, table in §4). Since $R \approx \ell_f$, the plate rigidity and buoyancy restoring force are both important. The maximum elastic deflection scales as

$$w_{\max} \sim \frac{q \, R^4}{D} = \frac{98 \times 10^3 \times (5000)^4}{8.7 \times 10^{17}} \approx 70 \text{ m}. \tag{38}$$

This crude scaling omits geometric prefactors that depend on the boundary conditions and load distribution. For a clamped circular plate, the central deflection is $q\,R^4/(64\,D) \approx 1.1$ m; the exact value for a floating plate lies between this and the full scaling estimate. The order of magnitude—decimeters to a meter—is consistent with GPS observations of elastic rebound following rapid subglacial lake drainage events in Antarctica [12].

6.3 Example 3: Supraglacial Lake Drainage and Ice-Shelf Hydrofracture

Meltwater lakes that form on the surface of ice shelves can trigger a catastrophic chain of events: the weight of the lake water flexes the ice shelf, producing tensile stresses at the base. If these stresses are large enough, they can initiate a fracture that propagates vertically through the full ice-shelf thickness—a process called **hydrofracture**. Hydrofracture has been proposed as the mechanism that caused the rapid disintegration of the Larsen B Ice Shelf in 2002 [11, 1].

Why elasticity? Lake loading and unloading occur on timescales of days to weeks (filling) and minutes to hours (drainage). The Deborah number for the drainage phase is $\mathrm{De} \gg 1$: the ice shelf responds elastically. The elastic analysis therefore governs whether hydrofracture can initiate.

Setup. Consider a circular supraglacial lake of radius R and depth d_w on an ice shelf of uniform thickness h. The lake applies a load $q = \rho_w \, g \, d_w$ to the ice-shelf surface (taking the lake water density as $\rho_w \approx 1000$ kg m^{-3}). The ice shelf floats on the ocean, so the flexure equation (12) governs the elastic deflection.

Flexural stress beneath the lake. The lake load causes the ice shelf to deflect downward beneath the lake and upward at its edges. The downward deflection produces a concave-up curvature, which puts the bottom of the ice shelf in tension. For a lake much smaller than the flexural wavelength ($R \ll \ell_f$), the maximum bending stress at the base of the ice shelf (at the center of the lake) scales as

$$\sigma_{\text{tens}} \sim \frac{3\,(1+\nu)\,\rho_w \, g \, d_w \, R^2}{8\,h^2}. \tag{39}$$

This is the maximum tensile stress at $z = -h/2$ beneath a uniform circular load on a thin plate, valid for $R \ll \ell_f$. The factor $(1 + \nu)$ arises from the biaxial bending moment at the center of an axisymmetric load.

Derivation: flexural stress beneath a circular lake load

For a uniform pressure q over a circle of radius R on an infinite plate floating on a fluid, the central deflection (for $R \ll \ell_f$) is approximately

$$w_0 \approx \frac{q\,R^4}{64\,D}.$$

The curvature at the center is $\kappa = d^2w/dr^2|_{r=0} \approx q\,R^2/(16\,D)$ (from the exact Kelvin function solution in the limit $R/\ell_f \to 0$). At the center of an axisymmetric problem, $\kappa_{rr} = \kappa_{\theta\theta} = \kappa$, so the bending stress at the bottom surface ($z = -h/2$) is

$$\sigma_{\text{tens}} = \frac{E\,h}{2(1-\nu^2)}\,(1+\nu)\,\kappa = \frac{E\,h\,(1+\nu)}{2(1-\nu^2)}\,\frac{q\,R^2}{16\,D} = \frac{E\,h}{2(1-\nu)}\,\frac{q\,R^2\,12(1-\nu^2)}{16\,E\,h^3} = \frac{3\,(1+\nu)\,q\,R^2}{8\,h^2}.$$

Substituting $q = \rho_w\,g\,d_w$:

$$\sigma_{\text{tens}} \approx \frac{3\,(1+\nu)\,\rho_w\,g\,d_w\,R^2}{8\,h^2}. \qquad \square$$

Numerical evaluation. Consider a lake with $d_w = 2$ m and $R = 500$ m on an ice shelf of thickness $h = 200$ m. The flexural wavelength is $\ell_f \approx 1.3$ km, so $R/\ell_f \approx 0.4$ and the small-load approximation is reasonable. Using $\nu = 0.325$:

$$\sigma_{\text{tens}} \approx \frac{3 \times 1.325 \times 1000 \times 9.8 \times 2 \times 500^2}{8 \times 200^2} = \frac{1.95 \times 10^{10}}{3.2 \times 10^5} \approx 61 \text{ kPa}. \tag{40}$$

This is about 6% of the tensile strength of intact ice (∼1 MPa). A deeper lake ($d_w = 5$ m) or a larger lake ($R = 1$ km) would produce stresses of 150 kPa or 240 kPa, respectively—still below the tensile strength of intact ice but potentially sufficient to reactivate pre-existing crevasses, which have a much lower critical stress intensity.

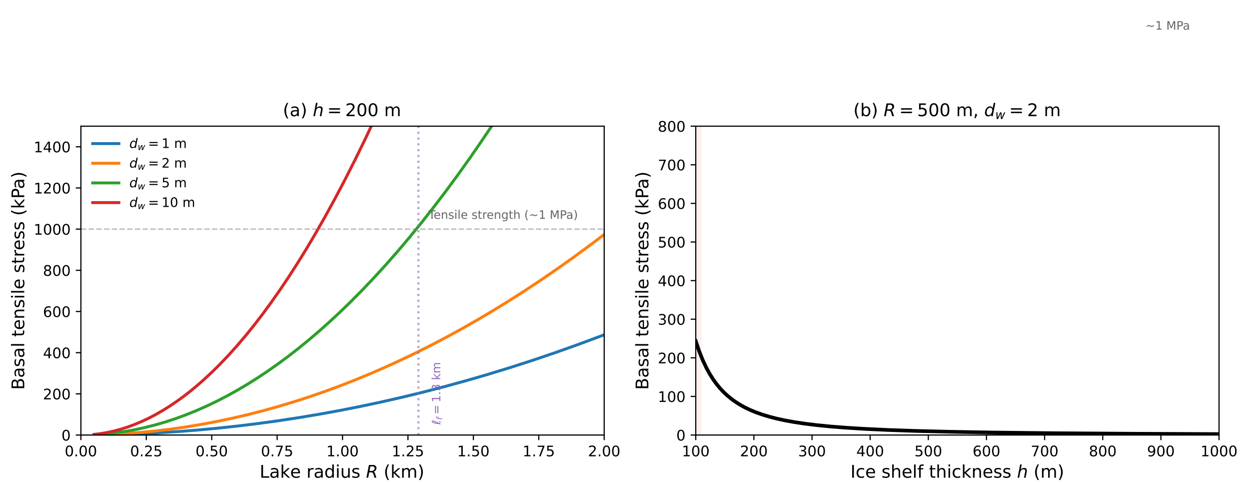


Figure 6: Basal tensile stress beneath a circular supraglacial lake from equation (39). (a) Stress as a function of lake radius for several lake depths on a 200 m thick ice shelf; the flexural wavelength ℓ_f is marked. (b) Stress as a function of ice-shelf thickness for a lake with $R = 500$ m and $d_w = 2$ m, showing that thin ice shelves are most vulnerable to hydrofracture initiation.

The hydrofracture feedback. Once a crevasse begins to propagate from the base (or surface) of the ice shelf, the water that enters the crack provides a pressure that drives the crack deeper. The key physics is that the hydrostatic pressure of water in the crack exceeds the lithostatic (ice overburden) pressure because $\rho_w > \rho_i$: at every depth, the water pressure in a water-filled crevasse exceeds the compressive stress from the weight of the overlying ice. This means a water-filled crevasse can, in principle, propagate through the full ice-shelf thickness regardless of thickness—a result first demonstrated by [15] and extended to ice shelves by [13, 11].

The elastic flexural analysis provides the initial stress state that determines whether hydrofracture can initiate. The subsequent propagation involves fracture mechanics (stress intensity factors, crack-tip processes) and the coupling between the fracture, the water supply, and the evolving stress field—topics beyond the scope of this lecture.

Connection to ice-shelf collapse. Prior to the disintegration of the Larsen B Ice Shelf in March 2002, satellite imagery showed extensive supraglacial lake formation on the shelf surface during the warm austral summer. Many of these lakes drained rapidly in the days immediately preceding the collapse. The elastic flexural stresses from the filling and draining of these lakes—compressive during filling, tensile during draining (from the elastic rebound)—are thought to have produced a "domino effect" in which the draining of one lake stressed neighboring regions, triggering further fracture and lake drainage [1]. This example illustrates how elastic processes operating on timescales of hours to days can trigger ice-shelf disintegration events that reshape the ice-sheet system over decades.

Reading

Cuffey & Paterson, *Physics of Glaciers*, 4th ed.:

- Section 3.2: Elasticity of ice
- Section 10.3: Flexure of floating ice
- Section 10.8: Calving and fracture

Petrenko & Whitworth, *Physics of Ice*, Chapter 4: Elastic properties.

A Hooke's Law from the First Law of Thermodynamics

In this appendix, we derive Hooke's law starting from the first law of thermodynamics for a deformable body under the assumption of adiabatic conditions (no heat exchange with the surroundings). The derivation proceeds in five steps: (1) write the first law for an adiabatic process; (2) express the rate of work done on the body in terms of surface tractions and body forces; (3) apply the divergence theorem; (4) identify Cauchy's equation of motion (the Stokes equation for a viscous fluid, or Newton's second law for a continuum) in the resulting expression; (5) define the strain energy density and deduce the form of the stiffness tensor.

Step 1: First law, adiabatic. The first law of thermodynamics states that the rate of change of total energy (internal plus kinetic) equals the rate of work done by external forces plus the rate of heat added:

$$\frac{d}{dt}(U + \mathcal{K}) = \mathcal{P} + \mathcal{Q}, \tag{41}$$

where U is the total internal energy, $\mathcal{K}$ is the kinetic energy, $\mathcal{P}$ is the rate of work by external forces, and $\mathcal{Q}$ is the rate of heat addition. For an adiabatic process, $\mathcal{Q} = 0$:

$$\frac{d}{dt}(U + \mathcal{K}) = \mathcal{P}. \tag{42}$$

Step 2: Rate of work. The external forces acting on a body occupying volume V with surface S are the surface tractions t_i and body forces f_i (per unit volume). The rate of work is

$$\mathcal{P} = \int_S t_i\, u_i\, dS + \int_V f_i\, u_i\, dV, \tag{43}$$

where $u_i = \dot{s}_i$ is the velocity.

Step 3: Divergence theorem. Using Cauchy's traction formula $t_i = \sigma_{ij}\, n_j$ (where n_j is the outward unit normal to S), the surface integral becomes

$$\int_S \sigma_{ij}\, u_i\, n_j\, dS = \int_V \frac{\partial}{\partial x_j}(\sigma_{ij}\, u_i)\, dV = \int_V \left(\frac{\partial \sigma_{ij}}{\partial x_j}\, u_i + \sigma_{ij}\, \frac{\partial u_i}{\partial x_j} \right) dV, \tag{44}$$

where we applied the divergence theorem in the first equality and the product rule in the second.

Step 4: Cauchy's equation of motion. Substituting (44) into (43):

$$\mathcal{P} = \int_V \underbrace{\left(\frac{\partial \sigma_{ij}}{\partial x_j} + f_i \right)}_{=\, \rho\, \dot{u}_i} u_i\, dV + \int_V \sigma_{ij}\, \frac{\partial u_i}{\partial x_j}\, dV. \tag{45}$$

The bracketed term is Cauchy's equation of motion (conservation of linear momentum):

$$\frac{\partial \sigma_{ij}}{\partial x_j} + f_i = \rho\, \dot{u}_i. \tag{46}$$

This is the continuum form of Newton's second law. In the Stokes equation for viscous flow, the inertial term $\rho\, \dot{u}_i$ is neglected (low Reynolds number), and the body force is gravity ($f_i = \rho\, g_i$); the form is the same.

The first integral in (45) is the rate of change of kinetic energy:

$$\int_V \rho\, \dot{u}_i\, u_i\, dV = \frac{d}{dt} \int_V \frac{1}{2} \rho\, u_i\, u_i\, dV = \frac{d\mathcal{K}}{dt}. \tag{47}$$

The second integral involves $\sigma_{ij}\,\partial u_i/\partial x_j$. Because σ_{ij} is symmetric, only the symmetric part of $\partial u_i/\partial x_j$ contributes:

$$\sigma_{ij}\,\frac{\partial u_i}{\partial x_j} = \sigma_{ij}\,\dot{\varepsilon}_{ij}, \tag{48}$$

where $\dot{\varepsilon}_{ij} = \frac{1}{2}(\partial u_i/\partial x_j + \partial u_j/\partial x_i)$ is the strain rate tensor.

Substituting into the first law (42):

$$\frac{dU}{dt} = \int_V \sigma_{ij}\,\dot{\varepsilon}_{ij}\,dV. \tag{49}$$

In terms of the internal energy per unit volume $\mathcal{U}$:

$$\boxed{\dot{\mathcal{U}} = \sigma_{ij}\,\dot{\varepsilon}_{ij},} \tag{50}$$

or equivalently, $d\mathcal{U} = \sigma_{ij}\,d\varepsilon_{ij}$. All of the mechanical work done on the body (minus the kinetic energy) is stored as internal energy—strain energy—in the material. This is the defining property of an elastic material: deformation is reversible and stores energy.

Step 5: Strain energy function and the stiffness tensor. Equation (50) implies that $\mathcal{U}$ is a function of the strain: $\mathcal{U} = \mathcal{U}(\varepsilon_{ij})$, and

$$\sigma_{ij} = \frac{\partial \mathcal{U}}{\partial \varepsilon_{ij}}. \tag{51}$$

A material for which such a strain energy function exists is called **hyperelastic**. For small strains, expand $\mathcal{U}$ as a Taylor series about the natural (stress-free) state $\varepsilon_{ij} = 0$:

$$\mathcal{U} = \underbrace{\mathcal{U}_0}_{=0} + \underbrace{\left.\frac{\partial \mathcal{U}}{\partial \varepsilon_{ij}}\right|_0}_{=\sigma_{ij}|_0 = 0} \varepsilon_{ij} + \frac{1}{2}\underbrace{\left.\frac{\partial^2 \mathcal{U}}{\partial \varepsilon_{ij}\,\partial \varepsilon_{kl}}\right|_0}_{=C_{ijkl}} \varepsilon_{ij}\,\varepsilon_{kl} + \cdots \tag{52}$$

The zeroth-order term is the energy of the reference state (set to zero). The first-order term vanishes because the reference state is stress-free. The leading nontrivial term is quadratic in strain, with coefficients

$$C_{ijkl} = \left.\frac{\partial^2 \mathcal{U}}{\partial \varepsilon_{ij}\,\partial \varepsilon_{kl}}\right|_0. \tag{53}$$

Differentiating:

$$\sigma_{ij} = \frac{\partial \mathcal{U}}{\partial \varepsilon_{ij}} = C_{ijkl}\,\varepsilon_{kl}. \tag{54}$$

This is Hooke's law (3). The stiffness tensor C_{ijkl} inherits the major symmetry $C_{ijkl} = C_{klij}$ from the equality of mixed partial derivatives of $\mathcal{U}$—a consequence of the existence of the strain energy function, which in turn follows from the adiabatic first law.

Isotropic reduction. For an isotropic material, C_{ijkl} must be invariant under all proper orthogonal transformations (rotations). The most general isotropic fourth-order tensor with the symmetries $C_{ijkl} = C_{jikl} = C_{ijlk} = C_{klij}$ is

$$C_{ijkl} = \lambda\,\delta_{ij}\,\delta_{kl} + \mu(\delta_{ik}\,\delta_{jl} + \delta_{il}\,\delta_{jk})\,, \tag{55}$$

giving

$$\sigma_{ij} = \lambda\,\varepsilon_{kk}\,\delta_{ij} + 2\mu\,\varepsilon_{ij}, \tag{56}$$

which is Hooke's law for an isotropic solid (5), with λ the first Lamé parameter and μ the shear modulus.

The strain energy density for an isotropic solid is therefore

$$\mathcal{U} = \frac{1}{2}\lambda\,(\varepsilon_{kk})^2 + \mu\,\varepsilon_{ij}\,\varepsilon_{ij} = \frac{1}{2}K\,(\varepsilon_{kk})^2 + \mu\,e_{ij}\,e_{ij}, \tag{57}$$

where the second form separates the volumetric (K) and deviatoric (μ) contributions. This confirms the physical interpretation: K controls the energy cost of volume change, and μ controls the energy cost of shape change.

B The Heat Equation from the First Law

In Appendix A, we assumed adiabatic conditions ($\mathcal{Q} = 0$) and obtained the elastic constitutive relation. Here, we take the complementary limit: allow heat flow but neglect mechanical work ($\mathcal{P} = 0$, or more precisely, neglect the strain-energy contribution). This yields the standard heat equation.

First law with heat flow. With no mechanical work, the first law (41) reduces to

$$\frac{dU}{dt} = \mathcal{Q}. \tag{58}$$

The rate of heat addition to a material volume V has two contributions: conduction through the surface S and internal heat generation at rate H per unit mass:

$$\mathcal{Q} = -\int_S q_i\, n_i\, dS + \int_V \rho\, H\, dV = \int_V \left(-\frac{\partial q_i}{\partial x_i} + \rho\, H \right) dV, \tag{59}$$

where q_i is the heat flux vector (positive in the direction of heat flow, hence the negative sign: heat flowing *out* through the surface reduces the internal energy) and H includes radiogenic heating or other volumetric sources. We have applied the divergence theorem to convert the surface integral to a volume integral.

Writing the internal energy per unit volume as $\mathcal{U} = \rho\, c_p\, (T - T_{\text{ref}})$ for a material with constant specific heat capacity c_p:

$$\rho\, c_p\, \frac{\partial T}{\partial t} = -\frac{\partial q_i}{\partial x_i} + \rho\, H. \tag{60}$$

Fourier's law. The constitutive relation for heat conduction is Fourier's law:

$$q_i = -k\, \frac{\partial T}{\partial x_i}, \tag{61}$$

where k is the thermal conductivity. For ice, $k \approx 2.1\ \mathrm{W\,m^{-1}\,K^{-1}}$ near 0°C (and increases at lower temperatures). Substituting:

$$\boxed{\rho\, c_p\, \frac{\partial T}{\partial t} = k\, \nabla^2 T + \rho\, H.} \tag{62}$$

This is the **standard heat equation**—a parabolic partial differential equation that governs the diffusion of temperature through a material in the absence of mechanical coupling.

The thermal diffusivity $\kappa = k/(\rho\, c_p)$ characterizes the rate of heat diffusion. For ice ($\rho = 917\ \mathrm{kg\,m^{-3}}$, $c_p = 2090\ \mathrm{J\,kg^{-1}\,K^{-1}}$, $k = 2.1\ \mathrm{W\,m^{-1}\,K^{-1}}$):

$$\kappa = \frac{2.1}{917 \times 2090} = 1.1 \times 10^{-6}\ \mathrm{m^2\,s^{-1}}. \tag{63}$$

The characteristic diffusion time over a length scale L is $t_{\text{diff}} \sim L^2/\kappa$. For an ice thickness of $h = 1000$ m: $t_{\text{diff}} \sim 10^6/(1.1 \times 10^{-6}) \approx 9 \times 10^{11}$ s $\approx 29\,000$ years. Temperature changes in the interior of a thick ice sheet diffuse only over timescales comparable to glacial cycles.

C The Generalized Heat Equation

Appendices A and B treated the two limits of the first law separately: adiabatic (no heat flow → elasticity) and no work (no deformation → heat equation). Here we retain both terms to obtain the **generalized heat equation**, which couples mechanical dissipation to the thermal field.

Full first law. Combining the results of the previous two appendices, the internal energy balance per unit volume is

$$\rho\, c_p \frac{\partial T}{\partial t} = \underbrace{\sigma_{ij}\,\dot\varepsilon_{ij}}_{\text{mechanical work}} + \underbrace{k\,\nabla^2 T}_{\text{heat conduction}} + \rho\, H. \tag{64}$$

The mechanical work term $\sigma_{ij}\,\dot\varepsilon_{ij}$ represents the rate at which stresses do work on the deforming material. In the adiabatic elastic case, this work is stored as recoverable strain energy. In a viscous fluid, this work is dissipated irreversibly as heat.

Viscous dissipation. For an incompressible viscous material ($\dot\varepsilon_{kk} = 0$, the condition appropriate for creeping glacier ice):

$$\sigma_{ij}\,\dot\varepsilon_{ij} = (-p\,\delta_{ij} + \tau_{ij})\,\dot\varepsilon_{ij} = -p\,\dot\varepsilon_{kk} + \tau_{ij}\,\dot\varepsilon_{ij} = \tau_{ij}\,\dot\varepsilon_{ij} \equiv \Phi, \tag{65}$$

where $\Phi = \tau_{ij}\,\dot\varepsilon_{ij}$ is the **viscous dissipation function**. Only the deviatoric stress does work; the pressure term vanishes because the flow is incompressible. Since both τ_{ij} and $\dot\varepsilon_{ij}$ are related by the flow law, Φ can be written in terms of the effective stress or effective strain rate alone:

$$\Phi = \tau_{ij}\,\dot\varepsilon_{ij} = 2\,\tau_e\,\dot\varepsilon_e = 2A\,\tau_e^{n+1}, \tag{66}$$

where the second equality uses $\dot\varepsilon_e = A\,\tau_e^n$. The dissipation is always positive ($\Phi > 0$).

Energy partition and the Taylor–Quinney coefficient. The mechanical work Φ does not all become heat. Some fraction is stored in the microstructure as the material deforms:

$$\Phi = \Phi_{\text{heat}} + \dot{\mathcal{U}}_{\text{disl}} + \dot{\mathcal{U}}_{\text{gb}}, \tag{67}$$

where $\dot{\mathcal{U}}_{\text{disl}}$ is the rate of elastic strain energy storage due to dislocations and $\dot{\mathcal{U}}_{\text{gb}}$ is the rate of surface energy change due to recrystallization.

Dislocation strain energy. Viscous creep in crystalline materials proceeds by the motion and multiplication of dislocations—line defects in the crystal lattice. Each dislocation carries an elastic strain field, and the associated elastic energy per unit length scales as $\sim \mu\, b^2$, where b is the magnitude of the Burgers vector ($b \approx 4.5 \times 10^{-10}$ m for basal slip in ice). For a dislocation density ϱ (total dislocation line length per unit volume), the stored elastic energy per unit volume is

$$\mathcal{U}_{\text{disl}} \sim \frac{1}{2}\,\mu\, b^2\, \varrho. \tag{68}$$

As deformation proceeds and ϱ increases, energy is drawn from the mechanical work and stored in the lattice rather than converted to heat.

Grain boundary surface energy. Grain boundaries carry a surface energy γ_{gb} per unit area ($\gamma_{\text{gb}} \approx 0.065\ \mathrm{J\,m^{-2}}$ for ice). If S_V denotes the grain boundary area per unit volume, the stored surface energy per unit volume is

$$\mathcal{U}_{\text{gb}} = \gamma_{\text{gb}}\, S_V. \tag{69}$$

During dynamic recrystallization, new grain boundaries nucleate and existing boundaries migrate, changing S_V. Nucleation of new grains increases S_V and stores energy; grain growth decreases S_V and releases energy as heat.

The **Taylor–Quinney coefficient** β is defined as the fraction of mechanical work that is converted to heat [8]:

$$\Phi_{\text{heat}} = \beta\,\Phi, \qquad 0 < \beta \leq 1. \tag{70}$$

The remaining fraction $(1-\beta)\,\Phi = \dot{\mathcal{U}}_{\text{disl}} + \dot{\mathcal{U}}_{\text{gb}}$ is stored in the microstructure.

Substituting into equation (64):

$$\boxed{\rho\, c_p\, \frac{\partial T}{\partial t} = 2\beta\, A\, \tau_e^{n+1} + k\,\nabla^2 T + \rho\, H.} \tag{71}$$

This is the **generalized heat equation for glacier ice**. In glaciology, β is almost universally taken to be unity: the energy stored in dislocations and grain boundaries is assumed negligible compared to the viscous dissipation. This is a reasonable approximation for steady-state creep, where dislocation production and annihilation are in approximate balance ($\dot{\varrho} \approx 0$) and the grain size has reached a statistical steady state ($\dot{S}_V \approx 0$), so that the stored energy terms vanish. During transient creep or active recrystallization, $\beta < 1$; measurements in metals undergoing plastic deformation give $\beta \approx 0.85$–0.95, though values for ice are not well constrained. Setting $\beta = 1$, the generalized heat equation reduces to

$$\rho\, c_p\, \frac{\partial T}{\partial t} = 2A\, \tau_e^{n+1} + k\,\nabla^2 T + \rho\, H, \tag{72}$$

which is the standard form used in ice-sheet models. The temperature changes due to three effects: viscous dissipation (first term), thermal conduction (second term), and internal heat generation (third term).

When does viscous dissipation matter? The ratio of viscous heating to conductive heat transport scales as

$$\frac{\Phi}{k\,\nabla^2 T} \sim \frac{2A\,\tau_e^{n+1}\,h^2}{k\,\Delta T}, \tag{73}$$

where h is the ice thickness and ΔT is the temperature difference across it. For typical ice-sheet conditions ($\tau_e = 0.1$ MPa, $A = 3.5 \times 10^{-25}$ $\mathrm{Pa^{-3}\,s^{-1}}$ at 263 K, $h = 1000$ m, $\Delta T = 30$ K, $k = 2.1\ \mathrm{W\,m^{-1}\,K^{-1}}$):

$$\frac{\Phi}{k\,\Delta T/h^2} = \frac{2 \times 3.5 \times 10^{-25} \times (10^5)^4 \times 10^6}{2.1 \times 30} = \frac{7 \times 10^{-25} \times 10^{26}}{63} = \frac{70}{63} \approx 1.1. \tag{74}$$

The two terms are comparable. Viscous dissipation is a significant heat source in the lower portions of thick, fast-flowing ice, where it warms the ice toward the pressure-melting point and softens it further—a positive feedback that concentrates deformation near the bed. In ice streams and shear margins, where τ_e is higher and the ice is warmer, viscous dissipation can be the dominant heat source.

The connection between elasticity and heat. The first law of thermodynamics provides a unified framework for understanding both elasticity and heat conduction. The adiabatic limit ($\mathcal{Q} = 0$) yields elasticity: mechanical work is stored reversibly as potential energy through the displacement of molecules from their equilibrium positions. The no-work limit ($\Phi = 0$) yields the heat equation: thermal energy, associated with the random vibrational motion of molecules, diffuses through the material. The full first law couples these two modes of energy: viscous dissipation converts ordered mechanical energy (the coherent translation and deformation of the material) into disordered thermal energy (the incoherent vibration of molecules about their lattice sites). Elasticity and heat are two manifestations of molecular-scale energy, distinguished by whether the molecular motion is ordered (elastic, recoverable) or disordered (thermal, irreversible).

D Derivation of the Thin Plate Flexure Equation

This appendix derives the governing equation for the bending of a thin elastic plate, equation (12), starting from the force balance on a plate element and the assumptions of Kirchhoff–Love plate theory. The derivation proceeds in four steps: (1) state the kinematic assumptions; (2) compute the internal stresses and bending moments; (3) derive the force and moment balance; (4) combine to obtain the biharmonic equation.

Step 1: Kinematic assumptions

Consider a thin plate of uniform thickness h, lying in the x–y plane. The plate's **neutral surface** (midplane) is at $z = 0$, so the plate extends from $z = -h/2$ to $z = h/2$. The key assumptions of **Kirchhoff–Love plate theory** are:

1. **Plane sections remain plane.** Cross sections that are initially normal to the midplane remain plane and normal to the deformed midplane after bending. This is the plate analogue of the Euler–Bernoulli beam assumption.

2. **No through-thickness stretching.** The plate thickness does not change during bending: $\varepsilon_{zz} = 0$.

3. **Small deflections.** The vertical deflection $w(x, y)$ of the midplane satisfies $|w| \ll h$ and $|\partial w/\partial x|, |\partial w/\partial y| \ll 1$, so that the geometry remains approximately flat.

Under these assumptions, the displacement field at any point (x, y, z) in the plate is determined entirely by the midplane deflection $w(x, y)$:

$$s_x(x, y, z) = -z\,\frac{\partial w}{\partial x}, \qquad s_y(x, y, z) = -z\,\frac{\partial w}{\partial y}, \qquad s_z(x, y, z) = w(x, y). \tag{75}$$

The first two expressions state that horizontal displacements arise from the rotation of the cross section: a point at height z above the midplane moves horizontally by $-z$ times the local slope.

Step 2: Strains, stresses, and bending moments

The in-plane strains follow from differentiating equation (75):

$$\varepsilon_{xx} = \frac{\partial s_x}{\partial x} = -z\,\frac{\partial^2 w}{\partial x^2}, \quad \varepsilon_{yy} = -z\,\frac{\partial^2 w}{\partial y^2}, \quad \varepsilon_{xy} = \frac{1}{2}\left(\frac{\partial s_x}{\partial y} + \frac{\partial s_y}{\partial x}\right) = -z\,\frac{\partial^2 w}{\partial x\,\partial y}. \tag{76}$$

All strains are proportional to z: the midplane ($z = 0$) is unstrained, and the strains increase linearly toward the surfaces. The curvatures of the midplane are $\kappa_{xx} = -\partial^2 w/\partial x^2$, $\kappa_{yy} = -\partial^2 w/\partial y^2$, and $\kappa_{xy} = -\partial^2 w/\partial x\,\partial y$, so that $\varepsilon_{ij} = z\,\kappa_{ij}$.

For a thin plate under bending, the through-thickness normal stress σ_{zz} is negligible compared to the in-plane stresses (this is the **plane stress** assumption: $\sigma_{zz} = 0$). Substituting the plane-stress form of Hooke's law:

$$\sigma_{xx} = \frac{E}{1-\nu^2}(\varepsilon_{xx} + \nu\,\varepsilon_{yy}) = -\frac{E\,z}{1-\nu^2}\left(\frac{\partial^2 w}{\partial x^2} + \nu\,\frac{\partial^2 w}{\partial y^2}\right), \tag{77}$$

and similarly for σ_{yy} and τ_{xy}. This is the bending stress formula stated in the main text, equation (16).

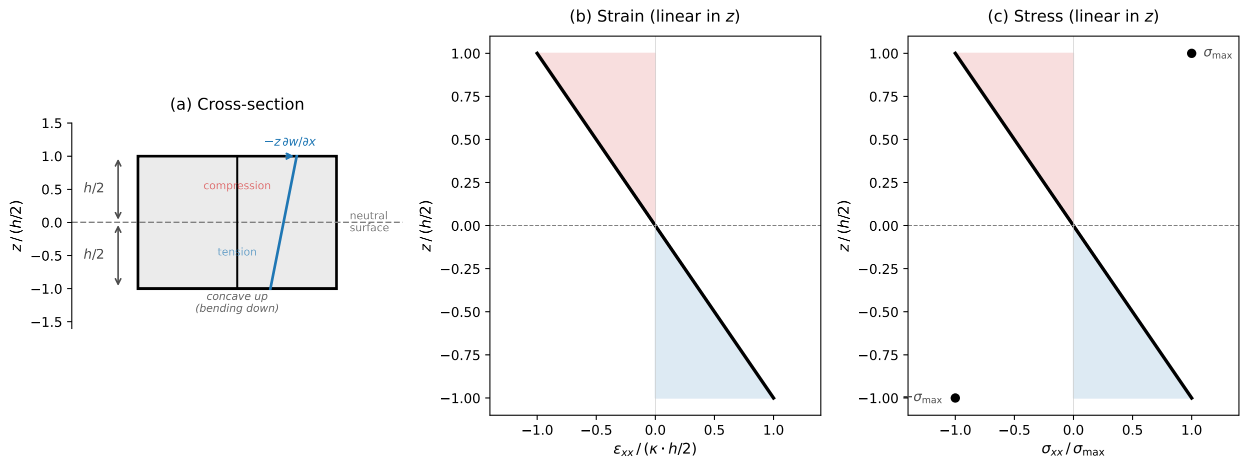


Figure 7: Elastic beam bending. (a) Cross-section of a beam of thickness h, showing the neutral surface ($z = 0$), the Kirchhoff–Love assumption (plane sections remain plane), and the horizontal displacement $-z\,\partial w/\partial x$ at height z. The upper half ($z > 0$) is in compression and the lower half ($z < 0$) in tension for a beam that is concave upward ($\kappa > 0$). (b) Strain ε_{xx} varies linearly through the thickness. (c) The resulting elastic bending stress also varies linearly, with maximum values $\pm\sigma_{\max}$ at the surfaces.

The **bending moments** per unit width are defined by integrating the stress times the moment arm z through the thickness:

$$M_{xx} = \int_{-h/2}^{h/2} \sigma_{xx}\, z\, dz, \qquad M_{yy} = \int_{-h/2}^{h/2} \sigma_{yy}\, z\, dz, \qquad M_{xy} = \int_{-h/2}^{h/2} \tau_{xy}\, z\, dz. \tag{78}$$

Substituting equation (77) and using $\int_{-h/2}^{h/2} z^2\, dz = h^3/12$:

$$M_{xx} = -D\left(\frac{\partial^2 w}{\partial x^2} + \nu\,\frac{\partial^2 w}{\partial y^2}\right), \qquad M_{yy} = -D\left(\frac{\partial^2 w}{\partial y^2} + \nu\,\frac{\partial^2 w}{\partial x^2}\right), \qquad M_{xy} = -D\,(1-\nu)\,\frac{\partial^2 w}{\partial x\,\partial y}, \tag{79}$$

where $D = E\,h^3/[12(1-\nu^2)]$ is the **flexural rigidity**, equation (11). The factor $h^3/12$ arises from the second moment of area of the plate cross section; the factor $(1-\nu^2)$ arises from the plane-stress constraint (the plate is free to contract laterally, unlike a plane-strain beam).

Step 3: Force and moment balance

Consider a small element of the plate with sides dx and dy. The element is subject to:

- **Applied loads**: a distributed transverse load $q(x, y)$ (force per unit area, positive upward) on the top surface, and a buoyancy restoring force $-\rho_w\, g\, w$ per unit area on the bottom surface (for a floating plate displaced by w from equilibrium).
- **Internal forces**: transverse shear forces Q_x and Q_y (force per unit length) acting on the cross-sectional faces.
- **Internal moments**: bending moments M_{xx}, M_{yy} and twisting moments M_{xy} acting on the faces.

The vertical force balance on the element gives:

$$\frac{\partial Q_x}{\partial x} + \frac{\partial Q_y}{\partial y} + q - \rho_w \, g \, w = 0. \tag{80}$$

The moment balance about the y-axis (taking moments on the element, neglecting higher-order terms) gives:

$$\frac{\partial M_{xx}}{\partial x} + \frac{\partial M_{xy}}{\partial y} = Q_x. \tag{81}$$

Similarly, the moment balance about the x-axis gives:

$$\frac{\partial M_{yy}}{\partial y} + \frac{\partial M_{xy}}{\partial x} = Q_y. \tag{82}$$

These two equations express the fact that the transverse shear forces are not independent quantities—they are determined by the spatial gradients of the bending and twisting moments.

Step 4: The general flexure equation for spatially varying D

Substitute equations (81) and (82) into the vertical force balance (80):

$$\frac{\partial^2 M_{xx}}{\partial x^2} + 2\,\frac{\partial^2 M_{xy}}{\partial x\,\partial y} + \frac{\partial^2 M_{yy}}{\partial y^2} + q - \rho_w \, g \, w = 0. \tag{83}$$

Now substitute the moment–curvature relations (79). Because the flexural rigidity $D = E\,h^3/[12(1-\nu^2)]$ depends on the local ice thickness $h(x, y)$, D must remain inside the derivatives. Computing each term with the product rule:

$$\begin{aligned}
\frac{\partial^2 M_{xx}}{\partial x^2} &= -\frac{\partial^2}{\partial x^2}\left[D\left(\frac{\partial^2 w}{\partial x^2} + \nu\,\frac{\partial^2 w}{\partial y^2}\right)\right], \\
2\,\frac{\partial^2 M_{xy}}{\partial x\,\partial y} &= -2\,\frac{\partial^2}{\partial x\,\partial y}\left[D\,(1-\nu)\,\frac{\partial^2 w}{\partial x\,\partial y}\right], \\
\frac{\partial^2 M_{yy}}{\partial y^2} &= -\frac{\partial^2}{\partial y^2}\left[D\left(\frac{\partial^2 w}{\partial y^2} + \nu\,\frac{\partial^2 w}{\partial x^2}\right)\right].
\end{aligned} \tag{84}$$

Substituting into equation (83) gives the **general flexure equation for a plate with spatially varying thickness**:

$$\boxed{\frac{\partial^2}{\partial x^2}\left[D\left(\frac{\partial^2 w}{\partial x^2} + \nu\,\frac{\partial^2 w}{\partial y^2}\right)\right] + 2\,\frac{\partial^2}{\partial x\,\partial y}\left[D\,(1-\nu)\,\frac{\partial^2 w}{\partial x\,\partial y}\right] + \frac{\partial^2}{\partial y^2}\left[D\left(\frac{\partial^2 w}{\partial y^2} + \nu\,\frac{\partial^2 w}{\partial x^2}\right)\right] + \rho_w \, g \, w = q.} \tag{85}$$

This is the form that should be used for ice shelves, where the thickness $h(x, y)$—and hence $D(x, y) \propto h^3$—varies significantly over the flexural wavelength. The cubic dependence on thickness means that a factor-of-two change in h produces an eightfold change in D, so even moderate thickness gradients generate substantial spatial variation in rigidity.

In one horizontal dimension (e.g., flow-line models of ice-shelf flexure), equation (85) simplifies to

$$\frac{d^2}{dx^2}\left(D\,\frac{d^2 w}{dx^2}\right) + \rho_w \, g \, w = q(x), \tag{86}$$

which, when expanded, contains terms involving dD/dx and d^2D/dx^2 that couple the thickness gradient to the deflection.

Constant-D limit. When D is spatially uniform (constant thickness), it factors out of the derivatives in equation (85). The three second-derivative terms then combine as:

$$D\left(\frac{\partial^4 w}{\partial x^4} + \nu\,\frac{\partial^4 w}{\partial x^2\,\partial y^2} + 2(1-\nu)\,\frac{\partial^4 w}{\partial x^2\,\partial y^2} + \frac{\partial^4 w}{\partial y^4} + \nu\,\frac{\partial^4 w}{\partial x^2\,\partial y^2}\right) = D\left(\frac{\partial^4 w}{\partial x^4} + 2\,\frac{\partial^4 w}{\partial x^2\,\partial y^2} + \frac{\partial^4 w}{\partial y^4}\right) = D\,\nabla^4 w, \tag{87}$$

where $\nabla^4 = \nabla^2(\nabla^2)$ is the biharmonic operator. The Poisson's ratio terms cancel: the coefficient of $\partial^4 w/(\partial x^2\,\partial y^2)$ is $\nu + 2(1-\nu) + \nu = 2$, independent of ν. The flexure equation reduces to

$$\boxed{D\,\nabla^4 w + \rho_w\,g\,w = q(x,y),} \tag{88}$$

which is equation (12). This biharmonic form is the classical plate equation of Kirchhoff (1850) and Love (1888). It is a useful approximation when the thickness varies slowly compared to the flexural wavelength ($|\nabla h/h| \ll 1/\ell_f$), but should be replaced by equation (85) when this condition is violated.

Boundary conditions

The flexure equation is fourth-order in each spatial coordinate, so it requires two boundary conditions at each edge. The standard boundary conditions are:

- **Clamped edge** (built-in): the deflection and slope are both zero,

$$w = 0, \qquad \frac{\partial w}{\partial n} = 0, \tag{89}$$

  where n is the coordinate normal to the edge. This is the appropriate condition at a grounding line where the ice is firmly attached to the bed (as in Example 1, §6.1).

- **Simply supported edge**: the deflection is zero but the plate is free to rotate,

$$w = 0, \qquad M_{nn} = 0, \tag{90}$$

  where M_{nn} is the bending moment about the edge. This condition sets the curvature (and hence the bending moment) to zero at the support.

- **Free edge**: both the bending moment and the effective transverse shear (the **Kirchhoff shear**, which combines the transverse shear force and the gradient of the twisting moment) vanish,

$$M_{nn} = 0, \qquad Q_n + \frac{\partial M_{nt}}{\partial t} = 0, \tag{91}$$

  where t is the coordinate along the edge. The second condition is Kirchhoff's boundary condition (1850): the twisting moment gradient acts as an equivalent transverse shear, so the plate edge must satisfy a combined force condition rather than separate conditions on Q_n and M_{nt}. This subtlety reduces the number of boundary conditions from three (which a fourth-order equation in 1D might seem to require at each end: M, Q, and M_{nt}) to two, consistent with the order of the equation.

For the tidal flexure problem (§6.1), the boundary conditions are: clamped at the grounding line ($w = 0$, $\partial w/\partial x = 0$ at $x = 0$) and bounded as $x \to \infty$. The boundedness condition replaces the two conditions at the far end and, together with the two clamped conditions, fully determines the solution.

E Viscous Thin Plate Bending

Appendix D derived the flexure equation for an elastic plate, where the bending moments are proportional to the curvature of the midplane. On timescales much longer than the Maxwell time ($\mathrm{De} \ll 1$), ice behaves as a viscous fluid. The elastic plate theory can be extended to a viscous fluid by replacing the elastic constitutive relation with a Newtonian viscous one: stress is proportional to strain *rate* rather than strain. The resulting equation governs the *rate* of bending rather than the bending itself, and is the appropriate description for slow flexural processes such as the viscous adjustment of grounding zones or the long-term sagging of ice shelves under sustained loads.

Constitutive relation for a thin Newtonian viscous plate

For a Newtonian incompressible viscous fluid, the deviatoric stress is $\tau_{ij} = 2\eta\,\dot{\varepsilon}_{ij}$, where η is the dynamic viscosity. The full Cauchy stress is $\sigma_{ij} = -p\,\delta_{ij} + 2\eta\,\dot{\varepsilon}_{ij}$ with the incompressibility constraint $\dot{\varepsilon}_{kk} = 0$.

Under the thin-plate (plane stress) assumption $\sigma_{zz} = 0$, the through-thickness strain rate is determined by incompressibility:

$$\sigma_{zz} = -p + 2\eta\,\dot{\varepsilon}_{zz} = 0 \quad \Longrightarrow \quad p = 2\eta\,\dot{\varepsilon}_{zz} = -2\eta\,(\dot{\varepsilon}_{xx} + \dot{\varepsilon}_{yy}), \tag{92}$$

where the last step uses incompressibility. Substituting back into the in-plane stresses:

$$\begin{aligned} \sigma_{xx} &= -p + 2\eta\,\dot{\varepsilon}_{xx} = 2\eta\,(2\dot{\varepsilon}_{xx} + \dot{\varepsilon}_{yy}), \\ \sigma_{yy} &= 2\eta\,(\dot{\varepsilon}_{xx} + 2\dot{\varepsilon}_{yy}), \\ \tau_{xy} &= 2\eta\,\dot{\varepsilon}_{xy}. \end{aligned} \tag{93}$$

These have the same structure as the elastic plane-stress relations (equation (77)) under the correspondence

$$\frac{E}{1-\nu^2} \;\longleftrightarrow\; 4\eta, \qquad \nu \;\longleftrightarrow\; \frac{1}{2}, \qquad \varepsilon_{ij} \;\longleftrightarrow\; \dot{\varepsilon}_{ij}. \tag{94}$$

The viscous "Poisson's ratio" is $\nu_v = 1/2$, which reflects the fact that an incompressible fluid preserves volume exactly under all deformations.

Kinematic assumption and strain rates

The Kirchhoff–Love kinematic assumption (Appendix D, Step 1) applies unchanged: plane sections normal to the midplane remain plane and normal after deformation. The velocity field at any point in the plate is

$$\dot{s}_x = -z\,\frac{\partial \dot{w}}{\partial x}, \qquad \dot{s}_y = -z\,\frac{\partial \dot{w}}{\partial y}, \qquad \dot{s}_z = \dot{w}(x,y,t), \tag{95}$$

where $\dot{w} = \partial w/\partial t$ is the rate of vertical deflection of the midplane. The in-plane strain rates are

$$\dot{\varepsilon}_{xx} = -z\,\frac{\partial^2 \dot{w}}{\partial x^2}, \qquad \dot{\varepsilon}_{yy} = -z\,\frac{\partial^2 \dot{w}}{\partial y^2}, \qquad \dot{\varepsilon}_{xy} = -z\,\frac{\partial^2 \dot{w}}{\partial x\,\partial y}. \tag{96}$$

Bending moments and the viscous flexural rigidity

The bending moments are defined as before (equation (78)). Substituting the viscous constitutive relation (93) with the strain rates (96) and integrating through the thickness:

$$\begin{aligned}
M_{xx} &= -D_v\left(\frac{\partial^2 \dot{w}}{\partial x^2} + \frac{1}{2}\frac{\partial^2 \dot{w}}{\partial y^2}\right), \\
M_{yy} &= -D_v\left(\frac{\partial^2 \dot{w}}{\partial y^2} + \frac{1}{2}\frac{\partial^2 \dot{w}}{\partial x^2}\right), \\
M_{xy} &= -D_v\,\frac{1}{2}\,\frac{\partial^2 \dot{w}}{\partial x\,\partial y},
\end{aligned} \tag{97}$$

where

$$\boxed{D_v = \frac{\eta\,h^3}{3}} \tag{98}$$

is the **viscous flexural rigidity**. This follows from the elastic expression $D = E\,h^3/[12(1-\nu^2)]$ under the substitution (94): $D_v = 4\eta\,h^3/12 = \eta\,h^3/3$. The factor 1/2 in the moment expressions is the viscous Poisson's ratio ν_v.

The viscous flexure equation

The force and moment balance equations (equations (80)–(82)) are statements of equilibrium and hold regardless of the constitutive law. Substituting the viscous bending moments (97) into the combined balance (83) and following the same algebraic steps as in Appendix D (the Poisson's ratio terms again cancel: $\frac{1}{2} + 2 \times \frac{1}{2} + \frac{1}{2} = 2$), we obtain the **viscous flexure equation**:

$$\boxed{D_v\,\nabla^4 \dot{w} + \rho_w\,g\,w = q(x, y, t).} \tag{99}$$

For spatially varying thickness, $D_v(x, y)$ must remain inside the derivatives, as in equation (85).

This equation is fundamentally different from the elastic flexure equation (88). The elastic equation relates the deflection w to the load q and is elliptic in space: given a load, the deflection is determined instantaneously. The viscous equation relates the *rate of deflection* $\dot{w}$ to the load and the current deflection, making it a parabolic PDE that is first-order in time and fourth-order in space. The deflection evolves: a suddenly applied load produces a gradual viscous sagging rather than an instantaneous elastic response.

Relaxation of a viscous floating plate

The character of equation (99) is best understood by considering the free relaxation of a floating plate after a load is removed. Setting $q = 0$ and decomposing w into spatial Fourier modes $w = \hat{w}(t)\,e^{ikx}$ in one dimension:

$$D_v\,k^4\,\dot{\hat{w}} + \rho_w\,g\,\hat{w} = 0 \quad\Longrightarrow\quad \hat{w}(t) = \hat{w}_0\,\exp\left(-\frac{t}{t_r}\right), \tag{100}$$

where the relaxation time for a mode of wavenumber k is

$$\boxed{t_r(k) = \frac{D_v\,k^4}{\rho_w\,g} = \frac{\eta\,h^3\,k^4}{3\,\rho_w\,g}.} \tag{101}$$

Short-wavelength perturbations (large k) relax *slowly* because the plate's viscous resistance to bending is greatest at short wavelengths. Long-wavelength perturbations (small k) relax quickly toward isostatic equilibrium because the plate offers little resistance to gentle bending.

The wavelength at which t_r equals a reference timescale t_0 defines a viscous flexural wavelength:

$$\ell_v = 2\pi\left(\frac{D_v}{\rho_w\, g\, t_0}\right)^{1/4} = 2\pi\left(\frac{\eta\, h^3}{3\,\rho_w\, g\, t_0}\right)^{1/4}. \tag{102}$$

Features with wavelengths shorter than ℓ_v are supported by the plate's viscous rigidity over the timescale t_0; features with wavelengths longer than ℓ_v have relaxed toward isostasy. Unlike the elastic flexural wavelength (14), which depends only on the plate properties, the viscous flexural wavelength depends on the observation timescale—a consequence of the time-dependent nature of viscous flow.

Steady-state harmonic response and the tidal phase lag

The relaxation analysis above describes the transient decay of an initial perturbation. For tidal flexure, we need the steady-state response to periodic forcing. Consider a one-dimensional viscous plate clamped at $x = 0$ (grounding line) and floating freely for $x > 0$, with tidal forcing $q = \rho_w\, g\, w_0\, \sin(\omega t)$. The viscous flexure equation (99) in 1D is

$$D_v\, \frac{\partial^4 \dot{w}}{\partial x^4} + \rho_w\, g\, w = \rho_w\, g\, w_0\, \sin(\omega t). \tag{103}$$

To find the steady-state response, write $w(x,t) = \mathrm{Im}\big[W(x)\, e^{i\omega t}\big]$, where $W(x)$ is a complex amplitude. Substituting and dividing by $e^{i\omega t}$:

$$i\omega\, D_v\, W'''' + \rho_w\, g\, W = \rho_w\, g\, w_0. \tag{104}$$

The particular solution is $W_p = w_0$ (constant). The homogeneous equation $W'''' = -\rho_w g/(i\omega D_v)\, W$ has characteristic roots $\lambda^4 = -\rho_w g/(i\omega D_v) = i\,\beta^4$, where

$$\beta = \left(\frac{\rho_w\, g}{\omega\, D_v}\right)^{1/4} \tag{105}$$

is the viscous flexural wavenumber. The four roots are $\lambda_j = \beta\, e^{i(\pi/8 + j\pi/2)}$ for $j = 0, 1, 2, 3$. Retaining only the two roots with $\mathrm{Re}(\lambda) < 0$ (so that $W \to w_0$ as $x \to \infty$):

$$\lambda_1 = \beta\, e^{i5\pi/8}, \qquad \lambda_2 = \beta\, e^{i9\pi/8}. \tag{106}$$

The general decaying solution is $W(x) = w_0 + C_1\, e^{\lambda_1 x} + C_2\, e^{\lambda_2 x}$, with C_1 and C_2 determined by the boundary conditions $W(0) = 0$ and $W'(0) = 0$ (clamped grounding line). Applying these:

$$C_1 = \frac{w_0\, \lambda_2}{\lambda_1 - \lambda_2}, \qquad C_2 = \frac{-w_0\, \lambda_1}{\lambda_1 - \lambda_2}. \tag{107}$$

The physical deflection $w(x,t) = \mathrm{Im}[W(x)\, e^{i\omega t}]$ can be written as

$$w(x,t) = |W(x)|\, \sin(\omega t - \varphi(x))\,, \tag{108}$$

where $|W(x)|$ is the local amplitude and $\varphi(x) = -\arg W(x)$ is the **phase lag** relative to the tidal forcing. In the far field, $W \to w_0$ (real and positive), so $\varphi \to 0$: the ice moves in phase with the tide. Near the grounding line, W is complex, and the phase lag is nonzero.

Physical interpretation. The parameter β controls both the spatial decay and the phase structure. The real and imaginary parts of the characteristic roots set the e-folding length and the spatial oscillation wavelength, both of order $1/\beta$. Comparing $1/\beta$ to the elastic flexural wavelength ℓ_f:

$$\frac{1}{\beta\,\ell_f} = \frac{1}{\ell_f}\left(\frac{\omega\,D_v}{\rho_w\,g}\right)^{1/4} = \left(\frac{\omega\,D_v}{4\,D}\right)^{1/4}. \tag{109}$$

This ratio is a Deborah-like number for flexure: when $\omega D_v \gg D$ (high viscosity or fast forcing), $1/\beta \gg \ell_f$ and the viscous adjustment zone is much wider than the elastic one, producing large phase lags. When $\omega D_v \ll D$, the viscous plate responds almost as quickly as an elastic one, and the phase lag is small.

Connection to the elastic and viscoelastic cases

The elastic and viscous flexure equations are the two limits of the Maxwell viscoelastic flexure equation. In a Maxwell material, the elastic and viscous curvature rates add in series. Writing this in bending-moment form (schematically, suppressing the Poisson coupling terms):

$$M + \frac{D_v}{D}\,\dot{M} = -D_v\,\dot{\kappa}, \tag{110}$$

where the ratio D_v/D plays the role of a relaxation time. Combining with the force balance $M'' + \rho_w\,g\,w = q$ yields a PDE that is first-order in time through both $\dot{w}$ and $\dot{M}$:

$$D_v\,\nabla^4\dot{w} + \rho_w\,g\,w + \frac{D_v}{D}\,\rho_w\,g\,\dot{w} = q + \frac{D_v}{D}\,\dot{q}. \tag{111}$$

- For De $\gg 1$ (fast loading, timescales much shorter than D_v/D): the time-derivative terms dominate. Dividing by D_v/D and integrating in time recovers the elastic flexure equation (88).
- For De $\ll 1$ (slow loading, timescales much longer than D_v/D): the time-derivative terms on the right-hand side and the $\dot{w}$ buoyancy term are negligible, and we recover the viscous flexure equation (99).

The ratio of the two rigidities is $D_v/D = \eta\,h^3/(3) \times 12(1-\nu^2)/(E\,h^3) = 4\eta(1-\nu^2)/E = 2\eta(1-\nu)/\mu$, using $E = 2\mu(1+\nu)$. For ice with $\nu = 0.325$: $D_v/D = 1.35\,\eta/\mu = 1.35\,t_M$. The Maxwell time thus controls the transition from elastic to viscous plate behavior, consistent with the bulk Deborah-number analysis in §5.3.

Extension to non-Newtonian (power-law) plate bending

The Newtonian viscous flexure equation is linear, which makes it analytically convenient but limits its applicability to glacier ice, where the creep law is nonlinear ($n \approx 3$–4). In this section we derive the moment–curvature-rate relation for a plate in cylindrical bending (deflection varying only in x) with a power-law rheology.

Setup. Consider a plate of thickness h undergoing cylindrical bending in the x–z plane, so that $\dot{\varepsilon}_{yy} = 0$ (the plate is wide compared to its flexural wavelength and the deflection does not vary in y). The Kirchhoff–Love kinematic assumption gives a strain rate that varies linearly through the thickness:

$$\dot{\varepsilon}_{xx}(z) = -z\,\dot{\kappa}, \qquad \dot{\kappa} \equiv \frac{\partial^2 \dot{w}}{\partial x^2}, \tag{112}$$

where $\dot\kappa$ is the rate of change of curvature. Incompressibility requires $\dot\varepsilon_{zz} = -\dot\varepsilon_{xx}$ (since $\dot\varepsilon_{yy} = 0$), and the plate is traction-free at the upper and lower surfaces ($\sigma_{zz} = 0$).

Stress state. From Glen's law $\dot\varepsilon_{ij} = A\,\tau_e^{n-1}\,\tau_{ij}$ with $\dot\varepsilon_{yy} = 0$:

$$\tau_{yy} = 0, \qquad \tau_{zz} = -\tau_{xx}, \qquad \tau_e = \left(\tfrac{1}{2}\tau_{ij}\tau_{ij}\right)^{1/2} = |\tau_{xx}|. \tag{113}$$

From $\sigma_{zz} = \tau_{zz} + p = 0$, the pressure is $p = -\tau_{zz} = \tau_{xx}$, so $\sigma_{xx} = \tau_{xx} + p = 2\,\tau_{xx}$ and $\tau_e = |\sigma_{xx}|/2$. Substituting into the flow law:

$$\dot\varepsilon_{xx} = A\,|\tau_{xx}|^{n-1}\,\tau_{xx} = A\left(\frac{|\sigma_{xx}|}{2}\right)^{n-1}\frac{\sigma_{xx}}{2} = \frac{A}{2^n}\,|\sigma_{xx}|^{n-1}\,\sigma_{xx}. \tag{114}$$

Inverting to express stress in terms of strain rate:

$$\sigma_{xx} = B\,|\dot\varepsilon_{xx}|^{(1/n)-1}\,\dot\varepsilon_{xx}, \qquad B \equiv \frac{2}{A^{1/n}}. \tag{115}$$

The constant B is the effective stiffness parameter for cylindrical bending, with units $\mathrm{Pa\,s}^{1/n}$. For $n = 1$, $B = 2/A = 4\eta$, recovering the Newtonian plane-strain result $\sigma_{xx} = 4\eta\,\dot\varepsilon_{xx}$ (consistent with the viscous flexural rigidity $D_v = \eta\,h^3/3$ derived in Appendix E).

Bending moment. The bending moment per unit width is

$$M = \int_{-h/2}^{h/2} \sigma_{xx}(z)\,z\,dz. \tag{116}$$

Substituting the strain rate (112) into the stress–strain-rate relation (115):

$$\sigma_{xx}(z) = B\,|z\,\dot\kappa|^{(1/n)-1}\,(-z\,\dot\kappa) = -B\,|\dot\kappa|^{(1/n)-1}\,\dot\kappa\,|z|^{(1/n)-1}\,z. \tag{117}$$

Note that $|z\,\dot\kappa|^{(1/n)-1} = |\dot\kappa|^{(1/n)-1}\,|z|^{(1/n)-1}$ because $1/n - 1 > -1$ for $n > 1$. Substituting into equation (116):

$$M = -B\,|\dot\kappa|^{(1/n)-1}\,\dot\kappa\int_{-h/2}^{h/2} |z|^{(1/n)-1}\,z^2\,dz. \tag{118}$$

The integrand $|z|^{(1/n)-1}\,z^2 = |z|^{(1/n)+1}$ is an even function of z, so

$$\int_{-h/2}^{h/2} |z|^{(1/n)+1}\,dz = 2\int_0^{h/2} z^{(1/n)+1}\,dz = 2\,\frac{z^{(1/n)+2}}{(1/n)+2}\bigg|_0^{h/2} = \frac{2}{1/n+2}\left(\frac{h}{2}\right)^{(1/n)+2} = \frac{2n}{1+2n}\left(\frac{h}{2}\right)^{(1/n)+2}. \tag{119}$$

Combining:

$$\boxed{M = -\frac{2n}{1+2n}\,B\left(\frac{h}{2}\right)^{(1/n)+2}\,|\dot\kappa|^{(1/n)-1}\,\dot\kappa.} \tag{120}$$

This is the **nonlinear moment–curvature-rate relation** for a power-law plate in cylindrical bending. It has the form $M = -D_n\,|\dot\kappa|^{(1/n)-1}\,\dot\kappa$, where the nonlinear flexural rigidity parameter is

$$D_n = \frac{2n}{1+2n}\,B\left(\frac{h}{2}\right)^{(1/n)+2} = \frac{4n}{1+2n}\,\frac{1}{A^{1/n}}\left(\frac{h}{2}\right)^{(1/n)+2}. \tag{121}$$

Newtonian check. For $n = 1$: $B = 4\eta = 2/A$, and $D_1 = \frac{2}{3}\cdot 4\eta\cdot\frac{h^3}{8} = \frac{\eta\,h^3}{3}$. This recovers the Newtonian viscous flexural rigidity $D_v = \eta\,h^3/3$ derived in Appendix E from the full plate equations—a necessary consistency check.

The nonlinear plate flexure equation. The 1D force balance (equation (83) in one dimension) is

$$\frac{d^2 M}{dx^2} + q - \rho_w\, g\, w = 0. \tag{122}$$

Substituting equation (120):

$$\boxed{\frac{d^2}{dx^2}\left[D_n \left|\frac{\partial^2 \dot{w}}{\partial x^2}\right|^{(1/n)-1} \frac{\partial^2 \dot{w}}{\partial x^2}\right] + \rho_w\, g\, w = q.} \tag{123}$$

This is a nonlinear parabolic PDE: fourth-order in x and first-order in t (through $\dot{w}$), with the nonlinearity entering through the power-law moment–curvature relation. For $n = 1$, it reduces to the linear Newtonian viscous flexure equation (99).

Key differences from the Newtonian case.

1. **Stress concentrates at the surfaces.** In a Newtonian plate, the bending stress varies linearly through the thickness ($\sigma_{xx} \propto z$). For $n > 1$, equation (117) gives $\sigma_{xx} \propto |z|^{1/n}$, which is sublinear: the stress is more uniform through the interior and rises steeply near the surfaces. This means that the surfaces carry a larger fraction of the bending moment than in the Newtonian case.

2. **The effective rigidity depends on the bending rate.** The moment scales as $M \propto |\dot{\kappa}|^{1/n}$ rather than $M \propto \dot{\kappa}$. Fast bending ($|\dot{\kappa}|$ large) produces a weaker-than-linear increase in the resisting moment, so the plate is effectively softer at high bending rates—the same strain-rate softening that characterizes power-law creep in simple shear.

3. **Superposition does not hold.** The flexure equation (123) is nonlinear, so solutions for different loads cannot be added. Each loading configuration must be solved independently, typically by numerical methods.

4. **The thickness dependence is weaker.** The flexural rigidity scales as $D_n \propto h^{(1/n)+2}$. For $n = 3$: $D_n \propto h^{7/3}$; for $n = 4$: $D_n \propto h^{9/4}$. Compare the Newtonian ($n = 1$) scaling $D_v \propto h^3$ and the elastic scaling $D \propto h^3$. The weaker thickness dependence for $n > 1$ means that thickness variations have a smaller effect on the nonlinear flexural rigidity than on the Newtonian or elastic rigidity—but the dependence is still strong enough that spatially varying h cannot be ignored.

Why the general 2D plate is harder. The cylindrical bending case derived above is tractable because the constraint $\dot{\varepsilon}_{yy} = 0$ reduces the problem to a single stress component. In a general 2D plate where both $\partial^2 \dot{w}/\partial x^2$ and $\partial^2 \dot{w}/\partial y^2$ are nonzero, the stress state is fully biaxial (σ_{xx}, σ_{yy}, τ_{xy} all nonzero), and the effective stress $\tau_e = (\frac{1}{2}\tau_{ij}\tau_{ij})^{1/2}$ couples all stress components. Inverting the flow law to express each stress component in terms of strain-rate components requires solving a nonlinear system at every point through the thickness and at every (x, y) location, which prevents the clean separation into independent moment–curvature relations that makes the cylindrical-bending and Newtonian cases tractable.

When does the nonlinearity matter? Critical tidal height

The numerical experiments in Figures 8 and 9 show that the power-law nonlinearity affects tidal flexure when the bending strain rate at the plate surface is comparable to or exceeds the background extensional strain rate $\dot{\varepsilon}_{\mathrm{bg}}$. We can derive a criterion for when nonlinearity becomes important.

The bending strain rate at the plate surface ($z = \pm h/2$) is $\dot{\varepsilon}_{\text{bend}} = (h/2)\,|\dot{w}''|$. The nonlinearity is significant when the tidal bending strain rate is comparable to the background:

$$\dot{\varepsilon}_{\text{bend}} = \frac{h}{2}\,|\dot{w}''| \sim \dot{\varepsilon}_{\text{bg}}. \tag{124}$$

From the elastic flexure solution (equation (34)), the peak curvature at the grounding line scales as $2w_0/\ell_f^2$, where $\ell_f = (4D/\rho_w g)^{1/4}$ is the flexural wavelength (equation (14)). The surface bending strain rate is therefore $\dot{\varepsilon}_{\text{bend}} \sim (h/2) \times (2w_0/\ell_f^2) \times \omega = h\,w_0\,\omega/\ell_f^2$. Setting this equal to $\dot{\varepsilon}_{\text{bg}}$:

$$\frac{h\,w_0\,\omega}{\ell_f^2} \sim \dot{\varepsilon}_{\text{bg}}. \tag{125}$$

Solving for the **critical tidal amplitude** above which nonlinearity matters:

$$\boxed{w_0^* = \frac{\dot{\varepsilon}_{\text{bg}}\,\ell_f^2}{h\,\omega}.} \tag{126}$$

For $w_0 \ll w_0^*$, the tidal perturbation is a small deviation from the background state and the linearized (Newtonian) response is adequate regardless of n. For $w_0 \gtrsim w_0^*$, the bending strain rate near the grounding line exceeds $\dot{\varepsilon}_{\text{bg}}$, and the effective viscosity becomes strain-rate dependent—the power-law nonlinearity shapes the flexure profile.

Numerical evaluation. For $\dot{\varepsilon}_{\text{bg}} = 10^{-10}$ s^{-1}, $h = 500$ m, $\ell_f = 2.6$ km, and $T_{\text{tide}} = 12$ hr ($\omega = 1.45 \times 10^{-4}$ s^{-1}):

$$w_0^* = \frac{10^{-10} \times (2600)^2}{500 \times 1.45 \times 10^{-4}} \approx 0.0093 \text{ m}. \tag{127}$$

This is much smaller than typical tidal amplitudes (0.5–2 m), indicating that the power-law nonlinearity is generally important for tidal flexure when background strain rates are as low as 10^{-10} s^{-1}. The Newtonian linearization becomes adequate only when $\dot{\varepsilon}_{\text{bg}}$ is large enough—for example, in fast-flowing ice shelves or shear margins where $\dot{\varepsilon}_{\text{bg}} \gtrsim 10^{-8}$ s^{-1}, the critical amplitude approaches 1 m and the linearization holds for typical tides. More generally, the nonlinearity matters whenever the tidal bending strains at the surface, of order $h\,w_0/\ell_f^2$, exceed the background strains accumulated over one tidal cycle, of order $\dot{\varepsilon}_{\text{bg}}/\omega$.

Multiple tidal constituents. Equation (126) applies to monochromatic (single-frequency) tidal forcing. In practice, two or more tidal constituents are often important at a given location (e.g., M_2 and S_2, or M_2 and O_1). The surface bending strain rate from multiple constituents is

$$\dot{\varepsilon}_{\text{bend}} \sim \frac{h}{\ell_f^2} \sum_j w_j\,\omega_j\,\sin(\omega_j\,t + \phi_j), \tag{128}$$

where w_j and ω_j are the amplitude and angular frequency of the jth constituent. The nonlinearity matters when the *total* bending strain rate exceeds $\dot{\varepsilon}_{\text{bg}}$, not when any single constituent does. The peak bending strain rate occurs during spring tides, when the constituent phases align constructively, giving the generalized criterion

$$\boxed{\frac{h}{\ell_f^2} \sum_j w_j\,\omega_j \lesssim \dot{\varepsilon}_{\text{bg}}.} \tag{129}$$

When the dominant constituents have similar frequencies (e.g., the semidiurnal M_2 and S_2), $\omega_j \approx \omega$ and equation (129) reduces to the monochromatic result (126) with w_0 replaced by the sum of the constituent amplitudes—the spring-tide amplitude. For mixed diurnal–semidiurnal tides, the frequency weighting means that higher-frequency constituents contribute disproportionately to the peak bending strain rate.

Effect of depth-dependent temperature

The derivations above assumed a uniform viscosity (or stiffness parameter B) through the thickness. In real glaciers, however, the temperature increases approximately linearly from the mean annual surface temperature T_s at $z = h/2$ to a value near the pressure melting temperature $T_b \approx 273$ K at $z = -h/2$:

$$T(z) = \bar{T} + G\,z, \qquad G = \frac{T_s - T_b}{h} < 0, \qquad \bar{T} = \frac{T_s + T_b}{2}. \tag{130}$$

Because the rate factor $A(T)$ follows an Arrhenius relation, the viscosity $\eta(z) = 1/[2A(T(z))]$ varies strongly with depth: the cold surface is stiff and the warm base is soft (Figure 11a). The surface-to-base viscosity ratio $\eta_s/\eta_b = A(T_b)/A(T_s)$ ranges from $\sim$10 for $T_s = 263$ K to several hundred for $T_s = 243$ K.

Newtonian case: shifted neutral axis. For cylindrical bending ($\dot{\varepsilon}_{yy} = 0$), the axial stress is $\sigma_{xx}(z) = 4\eta(z)\,\dot{\varepsilon}_{xx}(z)$ (the plane-strain stiffness, as derived in Appendix E). The Kirchhoff–Love kinematic assumption gives $\dot{\varepsilon}_{xx} = \dot{\varepsilon}_0 - z\,\dot{\kappa}$, where $\dot{\varepsilon}_0$ is the membrane strain rate and $\dot{\kappa} = \partial^2 \dot{w}/\partial x^2$. Requiring zero net axial force, $N = \int_{-h/2}^{h/2} \sigma_{xx}\,dz = 0$, determines the membrane strain rate:

$$\dot{\varepsilon}_0 = z_0\,\dot{\kappa}, \qquad z_0 \equiv \frac{\displaystyle\int_{-h/2}^{h/2} \eta(z)\,z\,dz}{\displaystyle\int_{-h/2}^{h/2} \eta(z)\,dz}. \tag{131}$$

The parameter z_0 is the viscosity-weighted centroid of the cross section; the strain rate vanishes at $z = z_0$, which defines the **neutral axis**. When η is uniform, $z_0 = 0$ and the neutral axis lies at the midplane. When the surface is stiffer than the base, $z_0 > 0$: the neutral axis shifts toward the cold, stiff surface.

The bending stress becomes $\sigma_{xx}(z) = -4\eta(z)\,(z - z_0)\,\dot{\kappa}$, and the bending moment $M = \int \sigma_{xx}\,z\,dz$ evaluates to

$$M = -\tilde{D}_v\,\dot{\kappa}, \qquad \boxed{\tilde{D}_v = 4\int_{-h/2}^{h/2} \eta(z)\,(z - z_0)^2\,dz.} \tag{132}$$

This is the **effective viscous flexural rigidity** for a plate with depth-dependent viscosity. The integral is the viscosity-weighted second moment of area about the neutral axis—a parallel-axis theorem. For uniform η, this reduces to $4\eta\,h^3/12 = \eta\,h^3/3 = D_v$, consistent with Appendix E. Because the force and moment balance equations are statements of equilibrium that hold regardless of the constitutive law, the viscous flexure equation (99) retains its form with $D_v \to \tilde{D}_v$. The relaxation times (101), flexural wavelength (102), and tidal phase-lag analysis all carry through with the modified rigidity.

Power-law extension. For a power-law rheology with depth-dependent stiffness $B(z) = 2/A(T(z))^{1/n}$ (equation (115)), the neutral-axis condition $N = 0$ becomes

$$\int_{-h/2}^{h/2} B(z)\,|z - z_0|^{(1/n)-1}\,(z - z_0)\,dz = 0. \tag{133}$$

The curvature rate $\dot{\kappa}$ cancels from this condition: the common factor $|\dot{\kappa}|^{1/n}$ divides out of the integrand. The neutral axis position z_0 therefore depends on n and $B(z)$ but *not* on the bending rate, which keeps the problem tractable. The nonlinear moment–curvature-rate relation becomes $M = -\tilde{D}_n\,|\dot{\kappa}|^{(1/n)-1}\,\dot{\kappa}$, where

$$\tilde{D}_n = \int_{-h/2}^{h/2} B(z)\,|z - z_0|^{(1/n)+1}\,dz. \tag{134}$$

For $n = 1$, equation (133) is linear in z_0 and recovers equation (131). For $n > 1$, equation (133) is a nonlinear equation that requires numerical root-finding, but this is a single scalar equation and straightforward to solve. The plate flexure equation (123) retains its structure with $D_n \to \tilde{D}_n$.

Physical effects (Figure 11). The depth-dependent temperature has two effects on viscous bending. First, the neutral axis shifts toward the cold, stiff surface ($z_0 > 0$). The cold upper portion of the plate, which carries most of the bending resistance, becomes thinner (measured from the neutral axis), while the warm lower portion contributes little. Second, the effective flexural rigidity $\tilde{D}_v$ is reduced relative to the value $D_v(\bar{T})$ evaluated at the depth-averaged temperature. Both effects make ice shelves more compliant than a depth-averaged viscosity would suggest. The rigidity reduction depends on the surface temperature and is typically 10–25% relative to $D_v(\bar{T})$ (Figure 11c), while the neutral axis can shift by more than half the distance from the midplane to the surface ($z_0 \sim 0.6\,h/2$ for $T_s = 243$ K). The stress asymmetry is the more consequential effect: the warm, soft base carries little bending stress, concentrating the load in the cold upper portion of the plate.

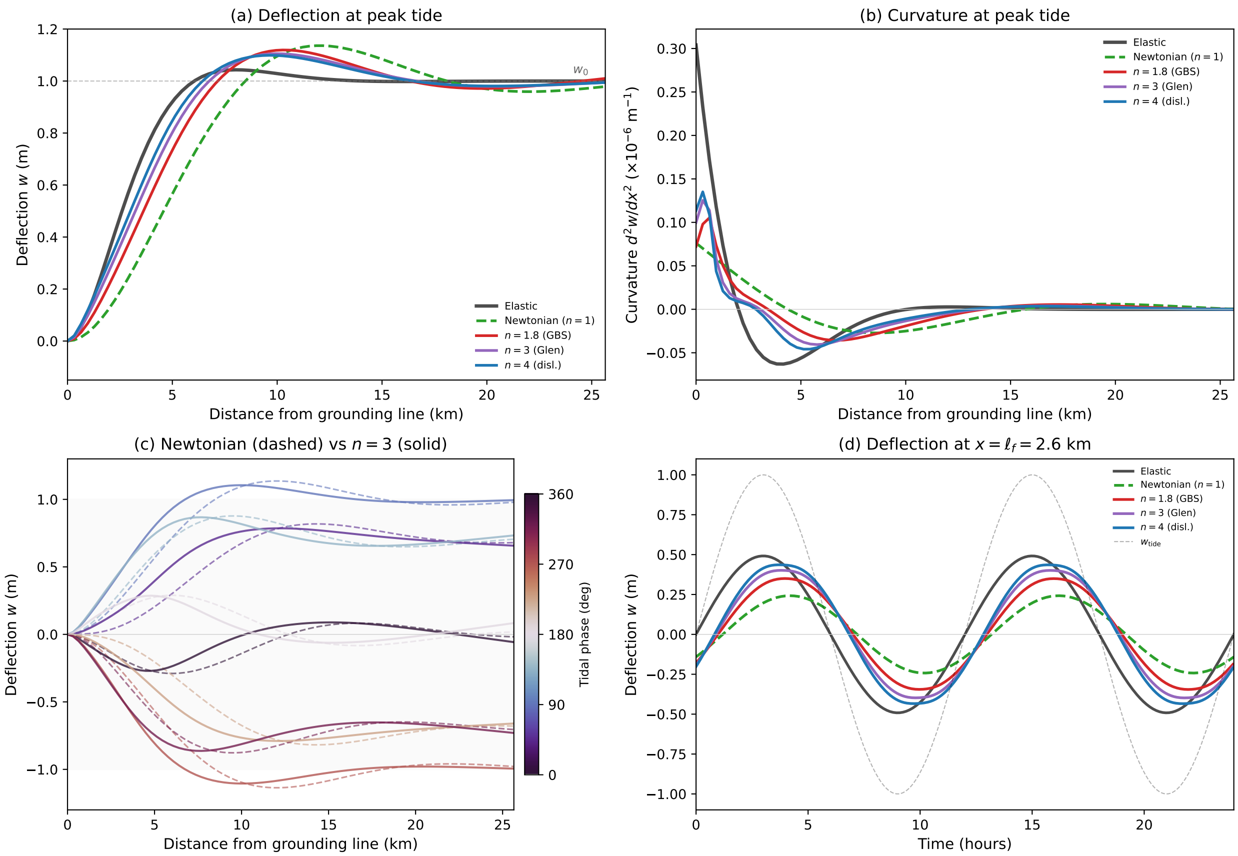


Figure 8: Tidal flexure at a grounding zone for five rheologies ($h = 500$ m, $w_0 = 1$ m, $T_{\text{tide}} = 12$ hr, $\dot{\varepsilon}_{\text{bg}} = 10^{-10}$ s^{-1}, $T = 263$ K). Flow law parameters from [5]: $n = 1.8$ (GBS-accommodated basal slip, $d = 1$ mm), $n = 3$ (Glen's law), $n = 4$ (dislocation creep). The Newtonian viscosity is set by Glen's law at $\tau_e = 0.1$ MPa. (a) Deflection profiles at peak tide. (b) Curvature profiles. (c) Snapshots at eight tidal phases for Newtonian (dashed) and $n = 3$ (solid). (d) Time series at $x = \ell_f$: the elastic response (dark grey) is in phase with the tide; all viscous responses lag the tide. Because the tidal bending strain rate $(h/2)\,|\dot{w}''| \gg \dot{\varepsilon}_{\text{bg}}$ (equation (126)), the power-law nonlinearity is active and the effective viscous rigidity D_{eff} exceeds D_v by a factor of 4–7. The power-law solutions are correspondingly stiffer and closer to the elastic response than the Newtonian solution. All three power-law flow laws remain nearly indistinguishable because they produce similar strain rates near the crossover stress of $\sim$0.1 MPa (§1).

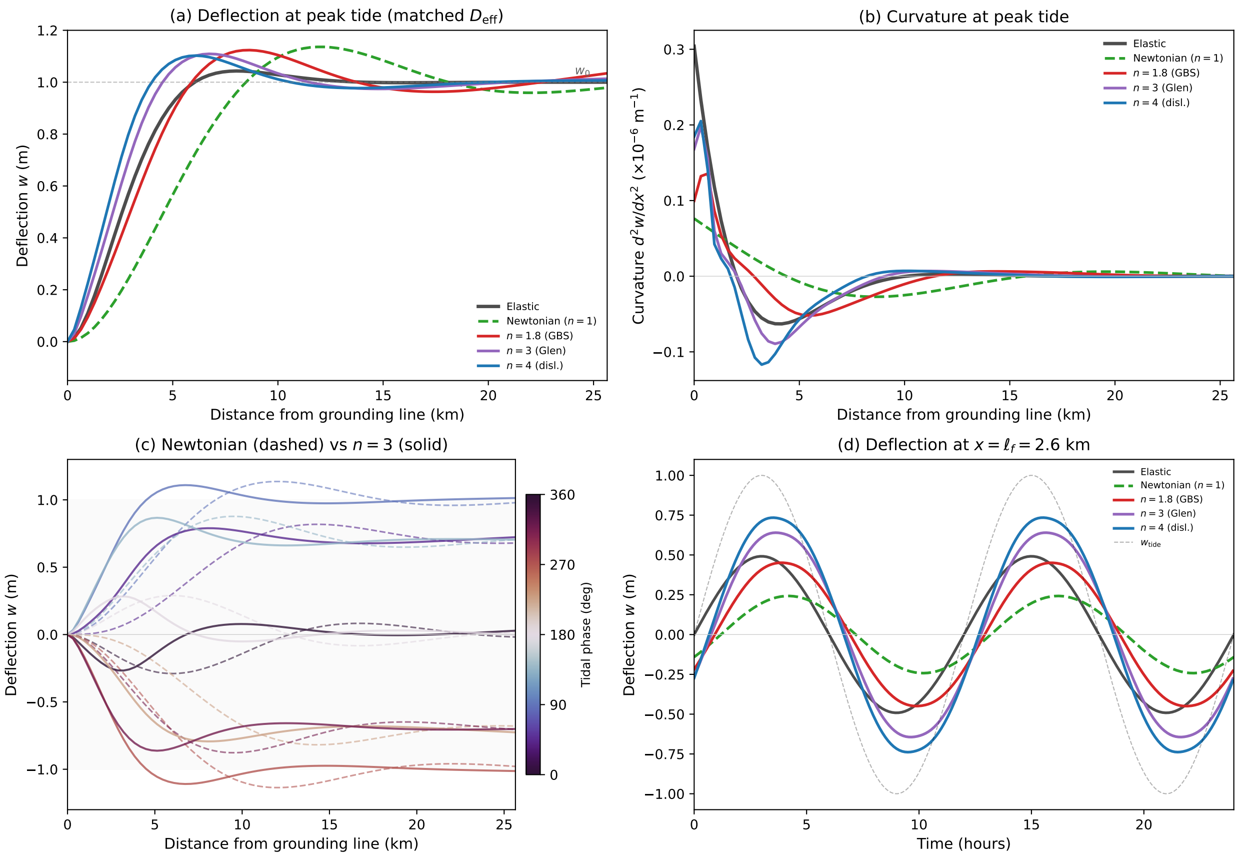


Figure 9: Same as Figure 8 but with all flow law prefactors matched so that $D_{\mathrm{eff}}(\dot{\kappa}_{\mathrm{bg}}) = D_v$, where $\dot{\kappa}_{\mathrm{bg}} = 2\dot{\varepsilon}_{\mathrm{bg}}/h$ is the background curvature rate. Even with matched background viscosities, the power-law solutions diverge from Newtonian because the tidal bending curvature rate exceeds $\dot{\kappa}_{\mathrm{bg}}$ near the grounding line ($w_0 \gg w_0^*$), causing strain-rate softening where bending is strongest.

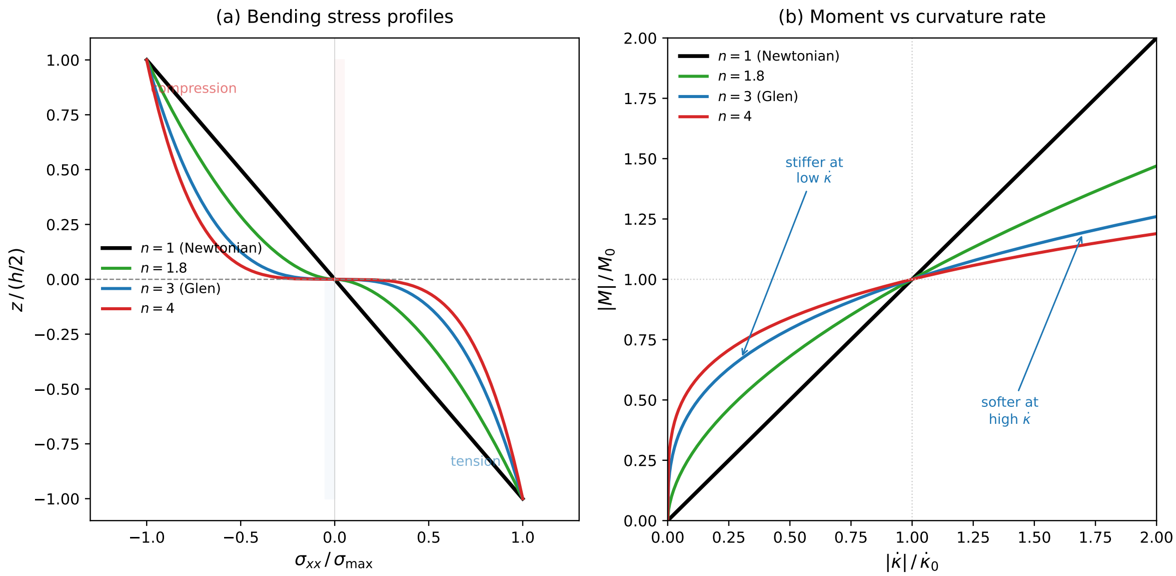


Figure 10: Viscous plate bending for Newtonian and power-law fluids. (a) Bending stress $\sigma_{xx}(z)$ through the plate thickness, normalized by the surface value. For $n = 1$ (Newtonian), the stress varies linearly in z; for $n > 1$, the stress varies as $|z|^{1/n}$, concentrating near the surfaces. (b) Bending moment $|M|$ as a function of curvature rate $|\dot{\kappa}|$. For $n = 1$, the relation is linear; for $n > 1$, the moment increases sublinearly ($M \propto |\dot{\kappa}|^{1/n}$), so the plate is effectively stiffer at low bending rates and softer at high bending rates.

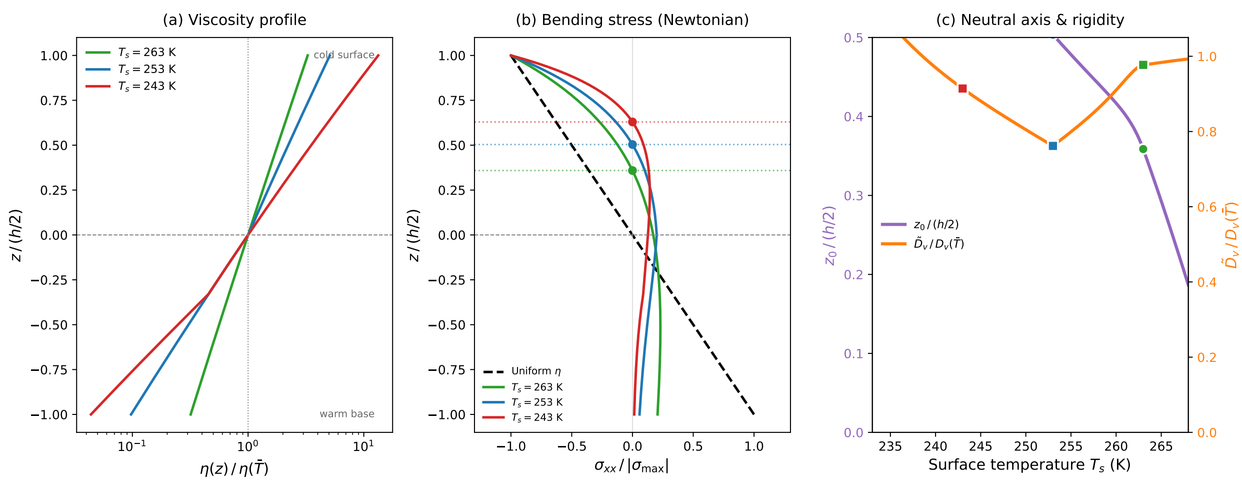


Figure 11: Effect of depth-dependent temperature on viscous plate bending, for a linear temperature profile with $T_b = 273$ K and varying surface temperature T_s. (a) Viscosity profile $\eta(z)/\eta(\bar{T})$: the surface-to-base contrast ranges from $\sim$10 ($T_s = 263$ K) to $\sim$300 ($T_s = 243$ K). (b) Newtonian bending stress $\sigma_{xx}(z)$ (normalized by the surface value) for uniform η (dashed black) and depth-dependent η (solid curves). The neutral axis (horizontal dotted lines) shifts toward the cold surface. (c) Neutral axis position $z_0/(h/2)$ and effective rigidity ratio $\tilde{D}_v/D_v(\bar{T})$ as functions of T_s. Colder surface temperatures produce larger asymmetry and greater rigidity reduction.

F Maxwell Viscoelastic Beam Bending

Appendix D derived the elastic flexure equation, in which the bending moment is proportional to the curvature: $M = -D\,\kappa$, where $\kappa = -\partial^2 w/\partial x^2$. Appendix E replaced this with a viscous relation, $M = -D_v\,\dot{\kappa}$ (where $\dot{\kappa} = -\partial^2 \dot{w}/\partial x^2$), appropriate for timescales much longer than the Maxwell time. The resulting viscoelastic flexure equation was stated in Appendix E (equation (111)); here we derive it from the moment–curvature relation and solve it analytically and numerically, valid for arbitrary Deborah number.

Viscoelastic moment–curvature relation

In a Maxwell material, the elastic and viscous curvature rates add in series. The bending moment obeys

$$M + t_r\,\dot{M} = -D_v\,\dot{\kappa}, \tag{135}$$

where $t_r \equiv D_v/D$ is the **flexural relaxation time**. This is the bending analogue of equation (21): the first term gives the viscous response ($M \propto \dot{\kappa}$) and the second gives the elastic response ($\dot{M} \propto \dot{\kappa}$, i.e. $M \propto \kappa$ upon integration).

The flexural relaxation time is related to the Maxwell time by

$$t_r = \frac{D_v}{D} = \frac{4\eta(1-\nu^2)}{E} = \frac{2\eta(1-\nu)}{\mu} = 2(1-\nu)\,t_M \approx 1.35\,t_M, \tag{136}$$

using $E = 2\mu(1+\nu)$, $t_M = \eta/\mu$, and $\nu = 0.325$ for ice (§3).

The viscoelastic flexure equation

The 1D force balance $M'' + \rho_w\,g\,w = q$ is a statement of equilibrium that holds regardless of the constitutive relation (here primes denote $\partial^2/\partial x^2$). Differentiating equation (135) twice in x gives $M'' + t_r\,\dot{M}'' = -D_v\,\dot{\kappa}''$. Substituting $\kappa = -w''$ and using $M'' = q - \rho_w g\,w$ and $\dot{M}'' = \dot{q} - \rho_w g\,\dot{w}$ from the force balance and its time derivative:

$$\boxed{D_v\,\frac{\partial^4 \dot{w}}{\partial x^4} + \rho_w\,g\,w + t_r\,\rho_w\,g\,\dot{w} = q + t_r\,\dot{q}.} \tag{137}$$

This is first-order in time and fourth-order in space, with the parameter t_r controlling the elastic–viscous transition. Recall that the Deborah number is $\mathrm{De} = t_M/t_p$, where t_p is the process timescale (§5.3); since $t_r \approx 1.35\,t_M$, the regime is set by the ratio $\omega\,t_r$:

- **Viscous limit** ($\mathrm{De} \ll 1$, $\omega\,t_r \ll 1$): The t_r terms are negligible, recovering the viscous flexure equation (99).
- **Elastic limit** ($\mathrm{De} \gg 1$, $\omega\,t_r \gg 1$): The t_r terms dominate. Dividing by t_r and integrating in time recovers $D\,w'''' + \rho_w g\,w = q$, the elastic flexure equation (88).

Analytical solution for Newtonian tidal forcing

For monochromatic tidal forcing $q = \rho_w\,g\,w_0\,\sin(\omega t)$, we seek the steady-state response $w(x,t) = \mathrm{Im}[W(x)\,e^{i\omega t}]$. Substituting into equation (137) and dividing by $e^{i\omega t}$:

$$i\omega\,D_v\,W'''' + \rho_w\,g\,(1 + i\omega\,t_r)\,W = \rho_w\,g\,(1 + i\omega\,t_r)\,w_0. \tag{138}$$

Define $\alpha \equiv 1 + i\omega\, t_r$. The particular solution is $W_p = w_0$, and the characteristic equation for the homogeneous part is

$$\lambda^4 = -\frac{\rho_w\, g\, \alpha}{i\omega\, D_v}. \tag{139}$$

Retaining the two roots with $\mathrm{Re}(\lambda) < 0$ (decaying as $x \to \infty$) and applying clamped boundary conditions $W(0) = 0$, $W'(0) = 0$ determines the solution. In the limits:

- $\omega\, t_r \to 0$ ($\mathrm{De} \to 0$): $\alpha \to 1$ and equation (138) reduces to the viscous harmonic ODE (104).
- $\omega\, t_r \to \infty$ ($\mathrm{De} \to \infty$): $\alpha \approx i\omega\, t_r$, so $-\rho_w g\, \alpha/(i\omega D_v) = -\rho_w g\, t_r/D_v = -\rho_w g/D$ and the characteristic equation becomes $\lambda^4 = -\rho_w g/D$, which is real. The four roots are the elastic flexural wavenumbers $\pm(1 \pm i)/\ell_f$, and the complex amplitude approaches $W(x) = w_0[1 - e^{-x/\ell_f}(\cos(x/\ell_f) + \sin(x/\ell_f))]$, the elastic solution (equation (34)).

Finite element discretization

For numerical solution—particularly when extending to nonlinear viscosity—we discretize equation (137) using Hermite cubic finite elements, the same basis used for the viscous solver in Appendix E. The weak form is: find $\dot{w} \in V_h$ such that for all $v \in V_h$,

$$D_v \int_0^L \dot{w}''\, v''\, dx + \rho_w g \int_0^L w\, v\, dx + t_r\, \rho_w g \int_0^L \dot{w}\, v\, dx = \int_0^L (q + t_r\, \dot{q})\, v\, dx. \tag{140}$$

With the semi-implicit treatment $w^{n+1} = w^n + \Delta t\, \dot{w}$ in the buoyancy term, this yields the linear system

$$\left[D_v\, \mathbf{K} + (\Delta t + t_r)\, \rho_w g\, \mathbf{M}\right] \dot{\mathbf{w}} = \rho_w g \left[(w_{\mathrm{tide}} - w^n) + t_r\, \dot{w}_{\mathrm{tide}}\right] \mathbf{f}, \tag{141}$$

where $\mathbf{K}$ is the bending stiffness matrix, $\mathbf{M}$ is the consistent mass matrix, and $\mathbf{f}$ is the load vector—all defined in terms of the Hermite shape functions.

Comparison to the pure viscous solver. The only modifications relative to the viscous FEM (Appendix E) are: (i) the mass matrix coefficient changes from Δt to $\Delta t + t_r$, and (ii) the right-hand side acquires the extra forcing term $t_r\, \rho_w g\, \dot{w}_{\mathrm{tide}}$. The stiffness matrix, boundary conditions, and time-stepping structure are unchanged. For nonlinear (power-law) viscosity, the Picard iteration on the effective viscous rigidity $D_{\mathrm{eff}}(\dot{\kappa})$ carries through exactly as in the viscous case; the elastic term $t_r\, \rho_w g\, \mathbf{M}$ is linear and does not participate in the iteration.

Results: Deborah number sweep

Figure 12 shows the viscoelastic tidal flexure solution for a 500-m-thick ice shelf across Deborah numbers spanning four orders of magnitude, from $\mathrm{De} = 0.01$ (viscous limit) to $\mathrm{De} = 100$ (elastic limit). As De increases:

1. The deflection profile sharpens from the broad, phase-shifted viscous response toward the narrow elastic flexure profile (panel a).
2. The peak curvature at the grounding line increases and the curvature pattern narrows (panel b).
3. The phase lag $\varphi(x)$ decreases monotonically toward zero—the ice responds increasingly in phase with the tide (panel c).

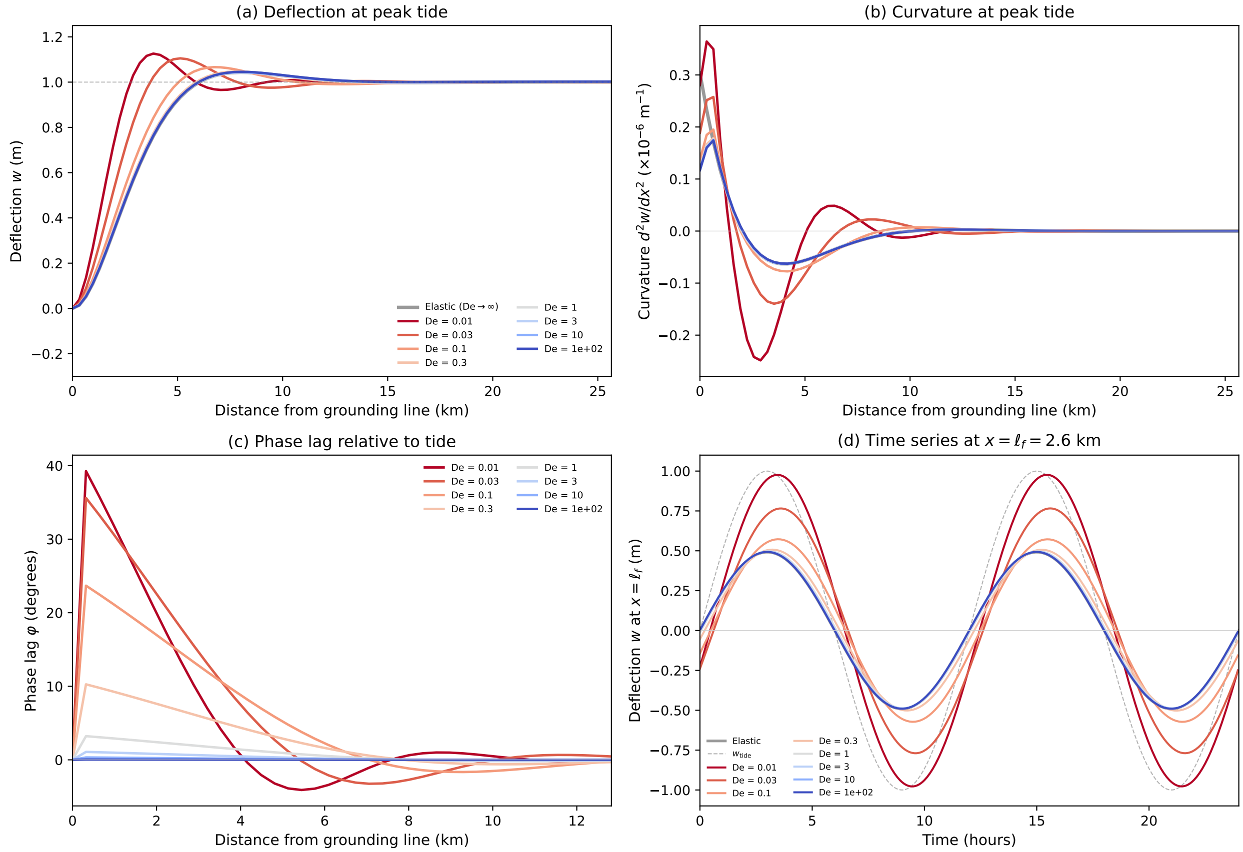


Figure 12: Maxwell (Newtonian) viscoelastic tidal flexure at a grounding zone ($h = 500$ m, $w_0 = 1$ m, $T_{\text{tide}} = 12$ hr), solved numerically for Deborah numbers from $\text{De} = 0.01$ (viscous limit, warm colors) to $\text{De} = 100$ (elastic limit, cool colors). (a) Deflection profiles at peak tide. (b) Curvature profiles. (c) Phase lag $\varphi(x)$ relative to the tidal forcing (from the analytical Newtonian solution). (d) Time series of deflection at $x = \ell_f$. The black curves show the elastic reference ($\text{De} \to \infty$). As De increases, the response transitions continuously from the broad, phase-shifted viscous solution to the narrow, in-phase elastic solution.

4. The time series at $x = \ell_f$ transitions continuously from the lagged, amplitude-reduced viscous response to the in-phase elastic response (panel d).

The Newtonian viscoelastic analytical solution provides a convenient summary of the De-dependent behavior (Figure 13). At $x = \ell_f$, the amplitude ratio $|W(\ell_f)|/w_0$ decreases monotonically from its viscous-limit value to the elastic value $1 - e^{-1}(\cos 1 + \sin 1) \approx 0.49$, while the phase lag decreases from its viscous-limit value to zero. The transition occurs over roughly $0.1 < \text{De} < 10$, centered near $\text{De} \approx 1$. For tidal periods of 12 hours, this corresponds to Maxwell times of ∼1–120 hours, or equivalently viscosities of $\sim 10^{13}$–10^{15} Pa s—precisely the range encountered in warm grounding zones (Table 2).

Effect of the flow law exponent. The Newtonian analytical results above provide a useful baseline, but glacier ice is a power-law fluid ($n \approx 1.8$–4). Figure 14 extends the Deborah number analysis to nonlinear viscosity by running the FEM solver for $n = 1$, 1.8, and 4, with the effective viscosity at the background curvature rate matched across all three flow laws at each De. The power-law nonlinearity has two effects. First, because the bending curvature rate near the grounding line

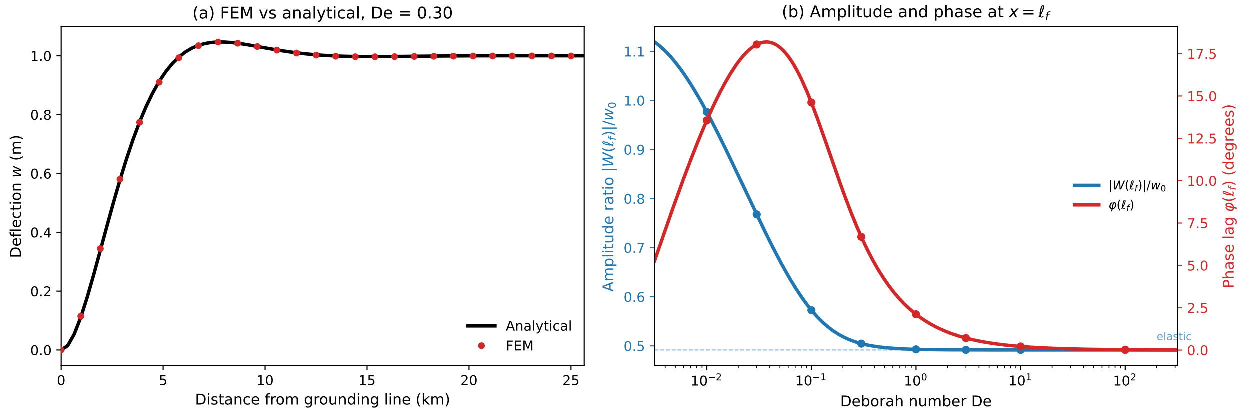


Figure 13: (a) FEM solution (circles) versus the analytical Newtonian viscoelastic solution (solid) at peak tide for De = 0.30. (b) Amplitude ratio $|W(\ell_f)|/w_0$ (blue) and phase lag $\varphi(\ell_f)$ (red) as continuous functions of De, evaluated from the analytical solution. Circles mark the FEM simulations from Figure 12. The dashed line in (b) marks the elastic-limit amplitude. The transition from viscous to elastic behavior spans roughly $0.1 < \text{De} < 10$.

exceeds the background rate (as in Figures 8–9), the effective viscosity there is reduced by strain-rate softening, making the power-law plate more compliant. This shifts the viscoelastic transition to lower De: the $n = 4$ response approaches the elastic limit at a smaller Deborah number than $n = 1$. Second, the amplitude and phase at $x = \ell_f$ show that higher n produces a faster transition between the viscous and elastic regimes, because the strain-rate-dependent viscosity amplifies the contrast between bending-dominated (near the grounding line) and far-field regions.
Figure 15 shows the full spatial and temporal structure of the power-law viscoelastic flexure, in the same four-panel format as the Newtonian sweep (Figure 12). At low De the three flow laws produce similar responses, but as De increases the strain-rate softening during bending gives each n a distinct effective De at the grounding line, producing visibly different flexure profiles, curvature fields, and phase lag structures. The differences are most pronounced in the curvature (panel b), where the power-law softening narrows and amplifies the curvature peak relative to the Newtonian solution at the same reference De.

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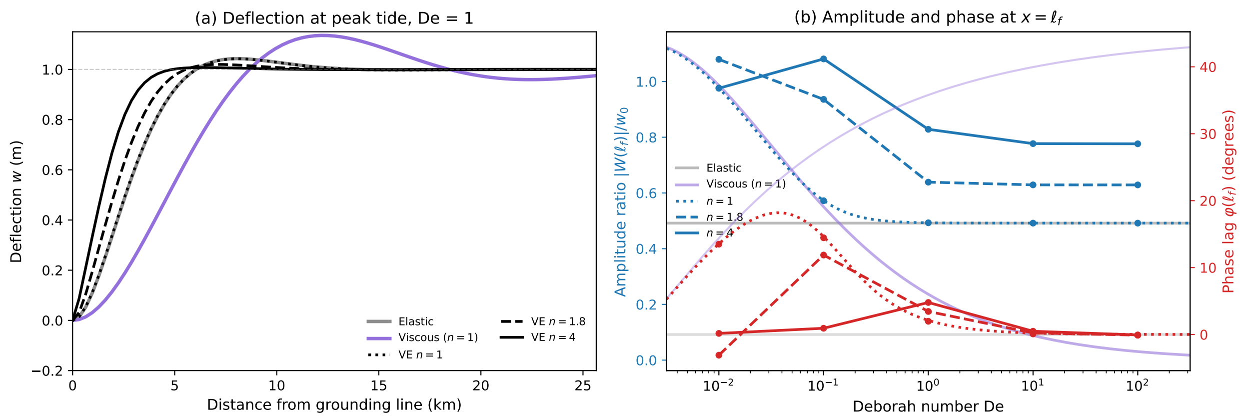

Figure 14: Effect of the flow law exponent on viscoelastic tidal flexure ($h = 500$ m, $w_0 = 1$ m, $T_{\text{tide}} = 12$ hr, $\dot{\varepsilon}_{\text{bg}} = 10^{-10}$ s^{-1}). Line styles: dotted ($n = 1$), dashed ($n = 1.8$), solid ($n = 4$). (a) Deflection at peak tide for De = 1, with the elastic (grey) and pure Newtonian viscous (purple) solutions for reference. (b) Amplitude ratio $|W(\ell_f)|/w_0$ (blue) and phase lag $\varphi(\ell_f)$ (red) as functions of De. Circles are FEM results; the $n = 1$ curves are the analytical Newtonian solution. The grey horizontal line marks the elastic limit; the purple curve is the pure viscous ($n = 1$, $D_{\text{el}} \to \infty$) solution parameterized by the same equivalent De.

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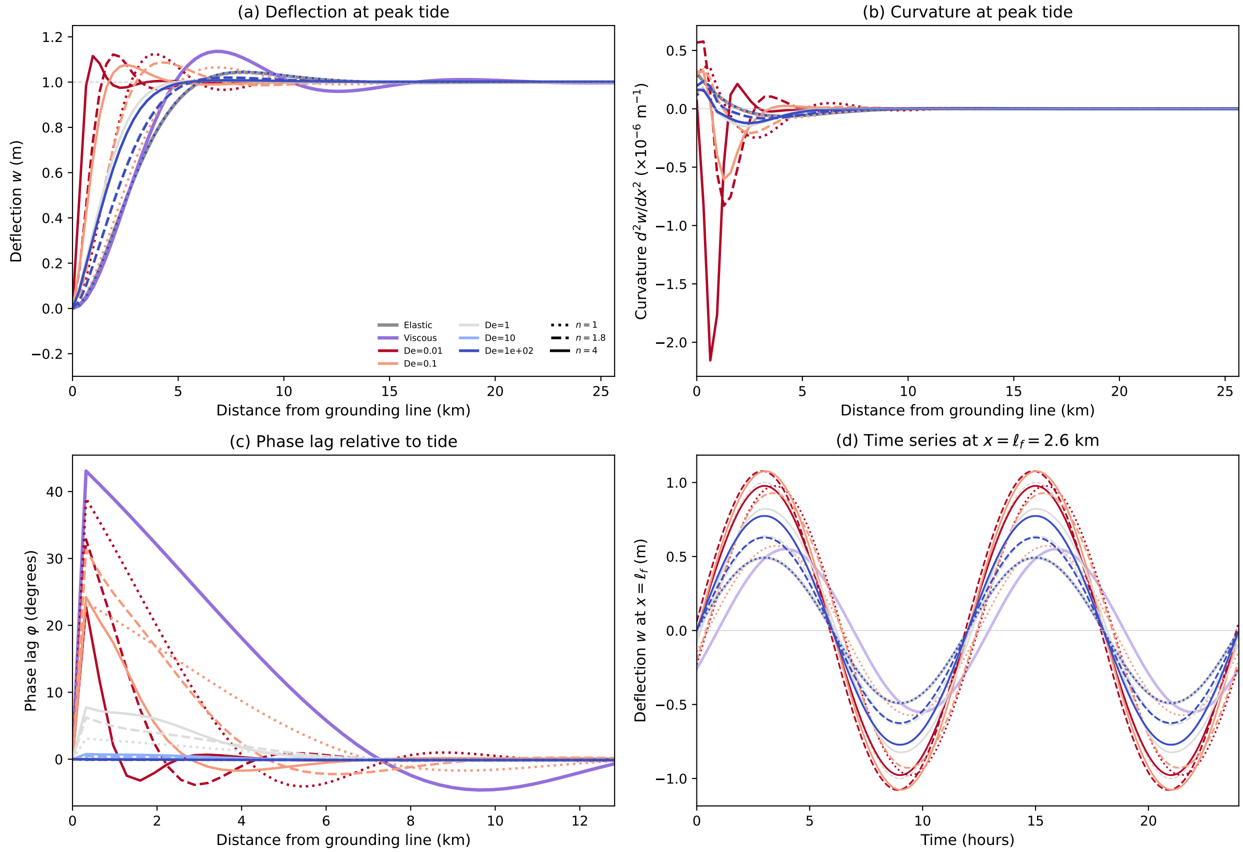

Figure 15: Power-law viscoelastic tidal flexure ($h = 500$ m, $w_0 = 1$ m, $T_{\text{tide}} = 12$ hr, $\dot{\varepsilon}_{\text{bg}} = 10^{-10}$ s^{-1}) for three flow law exponents: dotted ($n = 1$), dashed ($n = 1.8$), solid ($n = 4$), at five Deborah numbers from De = 0.01 (warm colors) to De = 100 (cool colors). Grey: elastic reference. Purple: pure Newtonian viscous reference at De = 0.1. (a) Deflection profiles at peak tide. (b) Curvature profiles. (c) Phase lag $\varphi(x)$ extracted from the FEM time history via Fourier analysis at each node. (d) Time series of deflection at $x = \ell_f$. The effective viscosity at the background curvature rate is matched across all three flow laws at each De, so differences arise from strain-rate softening during bending.

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